RIME CABIN

Owner-built mountain cabin · 24 x 32 ft · 2 bedrooms, 1 bath · 768 sq ft

Summit County, Colorado (Breckenridge area), about 9,500 to 10,000 ft. Lot not chosen yet.



Project data	
Use	Part-time single-family cabin, owner-occupied on visits. May be left unheated between trips.
Size	768 sq ft gross, one story with loft-ready ceilings. Rooms: living/kitchen/entry, 2 bedrooms, bath with laundry.
Construction	Type V-B wood frame. Concrete piers with flush LVL beams; hold-down piers under the braced end walls. 4:12 single-slope roof, metal.
Codes (assumed)	2021 IRC and 2021 IECC, climate zone 7, as adopted by the county. NFPA 58 for propane. NEC as adopted.
Design loads	Roof snow 100 psf (assumed), roof dead 21 psf, floor live 40 psf, wind 115 mph, Exposure C (Summit County lists 115 mph; C in open areas, B where treed), soil 2,000 psf presumptive (assumed; get a soils report).
Heat and fuel	Propane wall furnace plus wood stove. Propane tankless water heater and cooktop. 500 gal propane tank.
Power	Utility service, 100 A panel.
Water and waste	Private well (or hauled-water cistern) and septic system, designed separately.
Garage	Attached 1-car garage, 16 x 24 ft, on the west wall, door to the west, roof in the plane of the house roof. Not insulated; propane wall heater. See A-601.
Covered areas	North: house roof carried 7'-6" past the north wall to an LVL beam on four 6x6 posts, flush with the garage eave, about 8'-0 3/4" clear. West: garage roof carried south over the entry to the house's south eave, on an LVL beam and 3 8x8 posts; 8 x 12 ft 6 in entry deck under it. Open sides, cedar ceilings.
Estimate	About $231k with tax and a 10% contingency, garage included; site work, well, septic and utilities extra. See the parts list.

This design is for planning and permit preparation. Structural members, foundations, snow and wind loads, and all code items must be reviewed and stamped by a licensed engineer and approved by the local building department before construction.

SHEET INDEX

Sheet	Title
G-001	Cover, sheet index and project data
G-002	General notes, checks and specifications
C-101	Site plan
S-101	Foundation plan
S-102	Floor framing plan
S-201	Roof framing plans
S-301	Design criteria and member checks
S-401	Wind bracing plan
A-101	Floor plan
A-102	Roof plan
A-201	Exterior elevations
A-301	Building sections
A-401	Wall section and details
A-501	Schedules and envelope
A-601	Attached garage
P-101	Plumbing plan
P-201	Plumbing riser diagram and fixtures
E-101	Electrical plan
E-201	One-line diagram, panel schedule and load
M-101	Mechanical, ventilation and propane

CODE SUMMARY (CHECKED AGAINST THE MODEL, DETAILS ON G-002)

Result	Code and design check	Reference
PASS	Bedroom 1 size	IRC R304
PASS	Bedroom 1 egress window	IRC R310
PASS	Bedroom 2 size	IRC R304
PASS	Bedroom 2 egress window	IRC R310
PASS	Ceilings 9 ft or more	Your standard
PASS	Toilet clearances	IRC R307
PASS	Shower size	IRC P2708
PASS	Clear space at the shower door	IRC R307.1
PASS	Smoke and CO alarms	IRC R314, R315
PASS	GFCI and AFCI protection	NEC 210.8, 210.12
PASS	Outdoor receptacles	NEC 210.52(E)
PASS	Bath fan and fresh air	IRC M1505.4
PASS	Chimney height	IRC R1005, listing
PASS	Chimney air space	Chimney listing
PASS	Propane tank distance	NFPA 58
PASS	Vent and exhaust ends	IRC M1504.3, G2427.8
PASS	Garage separation from the house	IRC R302.5.1, R302.6, R315.2.2
PASS	Garage heater, gas and power	IRC G2408.2, NEC 110.26, 210.8
PASS	Covered area headroom	Planning check
PASS	Walls	IECC 2021 R402.1.3
WARN	Roof	IECC 2021 R402.1.3
PASS	Floor	IECC 2021 R402.1.3
PASS	Windows	IECC 2021 R402.1.2
PASS	No pipes in outside walls	Freeze protection
PASS	No pipes through beams	Structure
PASS	Members: rafters, joists, beams, headers, posts, studs	NDS, AISC, IRC R301
PASS	Footings and piers	IRC R403.1, IBC 1807
PASS	Wind bracing: shear walls, hold-downs, diaphragms	SDPWS, ASCE 7
ENGINEER	Engineer of record	IRC R301.1.3

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OWNER	Sander Blue
DRAWN BY	Claude (AI)
DATE	2026-10-10
SCALE	None

OPTIONS SHOWN

Concrete piers · 100 psf roof snow · Propane + wood · Utility power · 3 glass bays · Standing seam · No loft · Washer-dryer · Attached garage

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NOT FOR CONSTRUCTION

For planning and permit prep. Structural and code items need a licensed engineer and the building department.

Engineer's stamp

SHEET TITLE

COVER, SHEET INDEX AND PROJECT DATA

SHEET

G-001

GENERAL

1. These drawings are for planning and permit preparation. They are not for construction until reviewed and stamped by a licensed engineer and approved by the building department.
2. Design to the 2021 IRC and 2021 IECC unless the county has adopted a different year.
3. Verify all dimensions in the field. Dimensions are to the outside of framing unless noted.
4. The model, drawings and estimate come from one data file (model.json). Change the file, not the drawings.

FOUNDATION

1. Concrete 4,000 psi, air-entrained. Rebar ASTM A615 Grade 60.
2. Bottom of footings at least 48 in below grade or the county frost depth, whichever is deeper.
3. Piers: 12 in round (18 in under the covered-entry posts) with 4 #4 vertical bars, on footings sized from the loads. Shared pads and hold-down piers where marked on S-101.
4. Crawlspace: 20 x 10 in footing, 8 in stem wall, 1/2 in anchor bolts at 6 ft o.c., R-15 continuous insulation, sealed 12 mil vapor barrier, passive radon stack.

FRAMING

1. Exterior walls 2x6 at 16 in o.c. LSL studs where walls are taller than 12 ft (spacing on S-301).
2. Floor: 11 7/8 in I-joists (TJI 210 or equal) at 16 in o.c. in hangers between flush 3-ply LVL beams. LVL end beams under the east and west walls.
3. Roof: 16 in rafters on beveled plates, tied at every bearing. Simple spans that break over the north wall and the garage south wall. Spacing by snow load.
4. South glass wall: steel tube posts (HSS 5 x 3) between the glass bays with wood bucks; LVL headers sit on welded seats.
5. Bedroom wing ceiling: 2x10 at 16 in o.c., hung from a 2-ply 2x10 flush header near the north wall where the rafters come down; short 2x10s between the rafters north of it. Strong enough for a storage loft.
6. Wind bracing: shear wall panels, hold-downs, collectors and roof deck nailing per S-401.
7. Wall sheathing ZIP System R-6 (R-6.6 foam), all seams taped. It is the air and water barrier.

ENVELOPE

1. Walls R-23 mineral wool plus R-6.6 continuous (ZIP System R-6). Roof about R-60: 16 in of dense-pack cellulose (see the roof check). Floor R-44 dense-pack cellulose (piers) or R-15 crawl walls (crawlspace).
2. Windows U-0.30 or better.
3. Cold roof: a 3 1/2 in vent channel above the lower deck runs from the north soffit to the south soffit.
4. Air barrier: taped wall sheathing, taped lower roof deck, sealed subfloor. Target 3 air changes per hour at 50 Pa or better.

SYSTEMS

1. No water lines in exterior walls. All supply and drain lines run in the insulated floor north of the middle beam.
2. Low-point drain valve at the water entry. Blow out lines and add RV antifreeze to traps before leaving the house unheated.
3. Bath fan runs all the time at 50 cfm. That is the whole-house fresh-air system.
4. Smoke alarms in each bedroom, combined smoke/CO alarms in the living room near each bedroom door, all hardwired and interconnected.
5. Keep the roof clear of anything that sliding snow could hit except the chimney, which gets a snow splitter.

ASSUMPTIONS

1. Site in Summit County, Colorado (Breckenridge area), about 9,500 to 10,000 ft. Lot, road and utilities not chosen.
2. Road and driveway come from the west. The lot falls gently to the northeast.
3. Glass faces south within about 15 degrees.
4. Roof snow load is an input: 60, 100 or 150 psf. Your county sets the real number.
5. Wind 115 mph, Exposure C. Frost depth 48 in. Soil bearing 2,000 psf. All three need confirming.
6. Energy code: 2021 IECC, climate zone 7 (the toughest case in that elevation range).
7. The house can be left unheated between trips. All water lines are inside the insulation and can be drained.
8. Owner does most of the work. Prices are 2026 dollars with mountain delivery.
9. Footprint is fixed at 24 x 32 ft.

NOT VERIFIED

1. Member sizes for rafters, joists, beams, headers, posts, piers and the wind bracing. They come from the model's planning calculations (S-301, S-401), not an engineered design.
2. Your county's snow load, wind speed, frost depth and adopted code year.
3. Soil bearing. A soils test may change pier and footing sizes.
4. Window net clear openings. Check the egress size against the maker's data.
5. Wood stove clearances. Use the listing for the stove you buy.
6. Propane tank output on the coldest mornings, and whether to buy or lease the tank. Ask the propane company.
7. Prices marked Estimate in the parts list. They are my estimates; only lines marked as compared with a 2026 source were checked.
8. Whether a homeowner may do the plumbing, gas and electrical work in your county.

CHECKS RUN ON THE MODEL

Result	Check	What the model shows
PASS	Bedroom 1 size	86 sq ft, 9' 3" narrowest. Needs 70 sq ft and 7 ft.
PASS	Bedroom 1 egress window	E1: about 9.2 sq ft clear (31 x 43 in), sill 36 in. Needs 5.7 sq ft, 20 in wide, 24 in tall, sill 44 in or less. Confirm with the window maker.
PASS	Bedroom 2 size	75 sq ft, 8' narrowest. Needs 70 sq ft and 7 ft.
PASS	Bedroom 2 egress window	S4: about 9.2 sq ft clear (31 x 43 in), sill 36 in. Needs 5.7 sq ft, 20 in wide, 24 in tall, sill 44 in or less. Confirm with the window maker.
PASS	Ceilings 9 ft or more	Living room 9' 2" at the low wall, rising to 16' 10". Bedrooms and bath 9'.
PASS	Toilet clearances	16 in from the center to the nearest side obstruction (needs 15), 30 in clear in front (needs 21).
PASS	Shower size	36 x 34 in, 1224 sq in. Needs 30 in each way and 900 sq in. Specify a 22 in or wider door.
PASS	Clear space at the shower door	24 in clear between the vanity and the shower door (needs 24).
PASS	Smoke and CO alarms	Smoke alarm in each bedroom. 2 combined smoke/CO alarms in the living room by the bedroom doors. Hardwired and interconnected.
PASS	GFCI and AFCI protection	Checked device by device: every circuit with outlets in the kitchen, bath, laundry or outdoors has GFCI, and every circuit in the house has AFCI. Dual-function breakers on circuits 1, 2, 3, 4, 9. Heat cable on GFPE.
PASS	Outdoor receptacles	2 weatherproof GFCI receptacles, at the entry side and the back. Needs one at the front and one at the back.
PASS	Bath fan and fresh air	Bath fan runs at 50 cfm all the time. Whole-house need for 768 sq ft and 2 bedrooms is 45 cfm. 2021 IRC Table M1505.4.3(1): up to 1,500 sq ft with 0 to 1 bedrooms needs 30 cfm; 2 to 3 bedrooms needs 45 cfm.
PASS	Chimney height	Top is 5' 6" above the roof at the uphill side of the pipe (needs 3 ft) and 2' 2" above the highest roof within 10 ft (needs 2 ft). More than 5 ft above the roof, so it gets a brace.
PASS	Chimney air space	Roof opening 20 x 17 in (plan). The headers stay 2 in off the pipe at the top and bottom of the rafters, and a support box keeps the insulation back.
PASS	Propane tank distance	About 12' 2" from the siding to the 500 gal tank (needs 10 ft). The tank valves are 15' 11" from the water heater vent (keep 10 ft).
PASS	Vent and exhaust ends	None under the north eave, all 3 ft or more from windows and doors, and above the snow line (assumed 5' 9" of settled snow for 100 psf roof snow): furnace direct vent with snorkel kit 8' 1" above grade, water heater concentric vent 9' 7" above grade, bath fan duct to south wall cap 13' 2" above grade, range hood duct 11' 2" above grade. The stove air intake is under the main roof beside the entry deck.
PASS	Garage separation from the house	About 92 sq ft of 1/2 in gypsum on the garage side of the house wall, slab to roof deck. The door to the house is 1 3/4 in solid-core steel, self-closing and self-latching, and opens into the living area, not a bedroom. 2 carbon monoxide alarms in the house. The garage roof needs no rating because there is no room above it.
PASS	Garage heater, gas and power	Heater burner 19 in above the slab (needs 18). Vent ends 7' 6" above grade, above the assumed snow line. Gas comes off the house furnace line after the regulator, so no second regulator. Panel on the garage wall with open floor clear in front (needs 36 in) and 30 in wide. Garage circuit 13 on a GFCI breaker. Meter outdoors on the garage wall.
PASS	Covered area headroom	8' 1" clear under the beam and 8' 8" at the eave, above grade. Open to the north and east (the garage closes the west side).
PASS	Walls	As specified: R-23 mineral wool plus R-6.6 continuous foam (ZIP System R-6). Climate zone 7 needs R-20 plus R-5 continuous.
WARN	Roof	16 in of dense-pack cellulose is about R-59 to R-61 depending on density. Ask for 3.8 per inch, or use the IECC trade-off, to show R-60.
PASS	Floor	As specified: about R-44 dense-pack cellulose in 11 7/8 in bays. Zone 7 needs R-38.
PASS	Windows	Specified at U-0.30 or better.
PASS	No pipes in outside walls	Every supply and drain line is inside the insulation, so the house can sit unheated after it is drained.
PASS	No pipes through beams	All floor pipes stay north of the middle beam.
PASS	Members: rafters, joists, beams, headers, posts, studs	49 checks pass. Most used: floor joists: bending at 83% (11 7/8 in I-joist, TJI 210 or equal, at 16 in o.c., clear span 14' 8", 40 psf live, 12 psf dead, plus the bedroom partition and loft at 10').
PASS	Footings and piers	15 checks pass. Most used: covered-area post footings: wind uplift at 100% (About 21 psf net on the open roof; 1,372 lb at the worst post, held by its footing and the soil over it. Post base and beam cap rated at least this much).
PASS	Wind bracing: shear walls, hold-downs, diaphragms	38 checks pass. Most used: wind bracing, garage west wall at 100% (GW-B: 1 panel, 4' 8" effective, up to 862 plf).
ENG.	Engineer of record	Rafters: 16 in heavy-flange I-joist at 12 in o.c. Sizes come from the model's own load calculations for 100 psf roof snow and 115 mph wind (S-301, S-401). A Colorado-licensed engineer checks them and stamps the drawings.
PASS	Every part is priced	All 2,476 model parts map to a parts-list line or a site allowance, and every line maps to parts or a stated allowance.
PASS	Areas add up	Rooms 696 + outside walls 50 + inside walls 14 + drywall about 8 = 768 sq ft gross.

ABBREVIATIONS

Abbreviation	Meaning
AFF	Above finished floor
C/L	Center line
AAV	Air admittance valve (lets air into a drain without a vent pipe)
ACH50	Air changes per hour at a 50 pascal blower-door test
ci	Continuous insulation (foam over the studs)
CSST	Corrugated stainless steel tubing (gas)
DWV	Drain, waste and vent
GFCI / AFCI	Ground-fault / arc-fault protection. GFCI/AFCI = one dual-function breaker
GFPE	Ground-fault protection of equipment (heat cable breaker)
LVL / LSL	Laminated veneer / strand lumber
O.C.	On center
PT	Pressure-treated
T.O.	Top of
ZIP-R	Insulated structural sheathing

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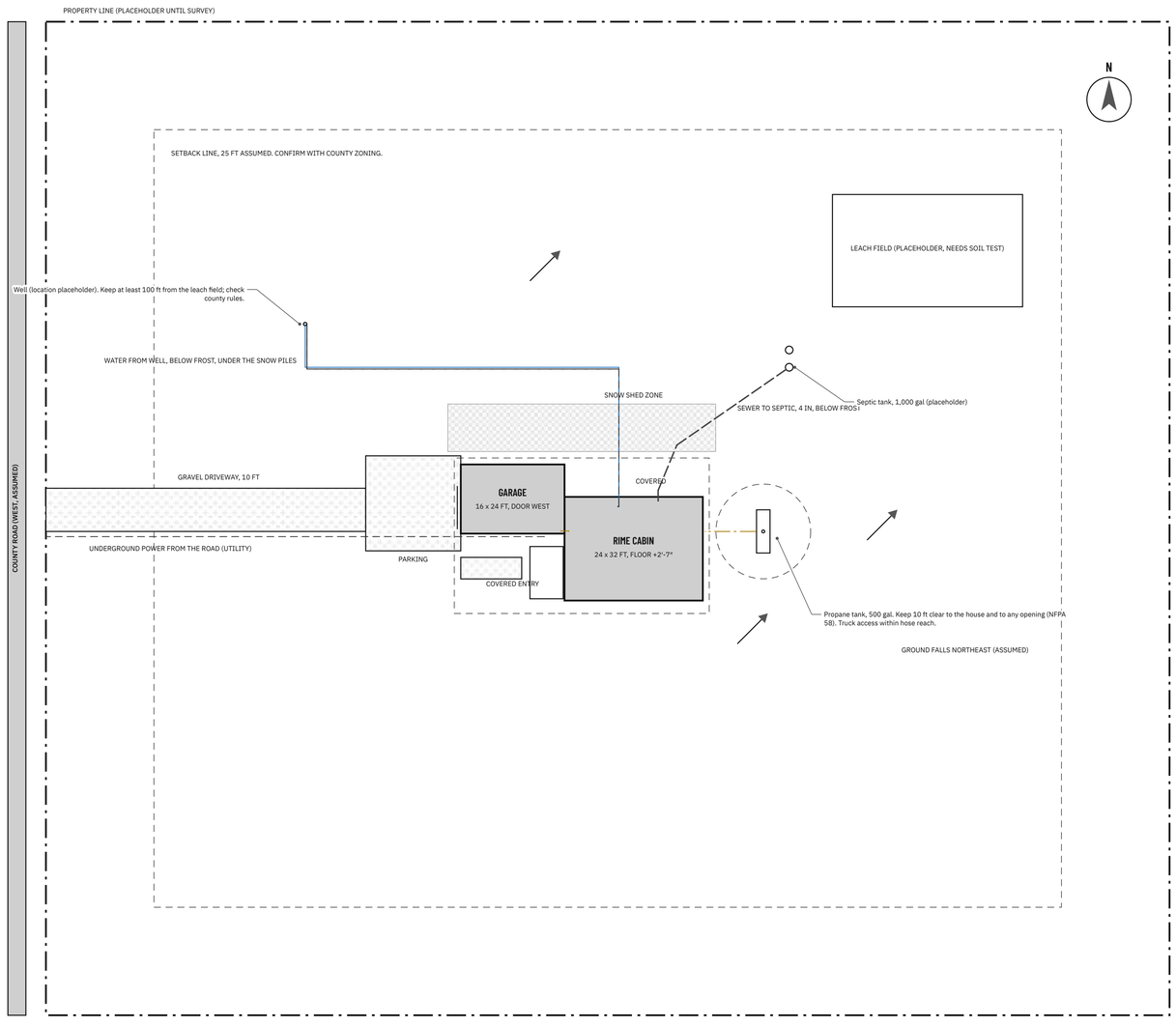
Engineer's stamp

SHEET TITLE

GENERAL NOTES, CHECKS AND SPECIFICATIONS

SHEET

G-002



1 **SITE PLAN**

1/16" = 1'-0" All locations are placeholders until a lot is chosen and surveyed

SITE NOTES

1. This plan is a placeholder layout. Pick the lot, then survey it and re-site the house before the permit.
2. Face the glass within about 15 degrees of true south. Keep trees to the south low or thin them for winter sun.
3. Road and driveway from the west. Keep the driveway, propane truck route and parking out of the north snow shed zones.
4. Septic: soil test (perc test) first. The design comes from a licensed installer or engineer and the county health department.
5. Well: drill before building if possible. If the well is poor, a hauled-water cistern is the fallback (see the parts list).
6. Keep the well at least 100 ft from the leach field and 50 ft from the septic tank, or what the county requires.
7. Power: ask the utility for a line extension quote early. Off-grid solar is the alternative in the cost section.
8. Grade away from the house at least 6 in in the first 10 ft. Keep 8 in between grade and siding.
9. Garage (if built): attached on the north half of the west (entry) wall, door west to the parking. Its roof continues the house roof and sheds snow north, and runs on south over the entry. See A-601.

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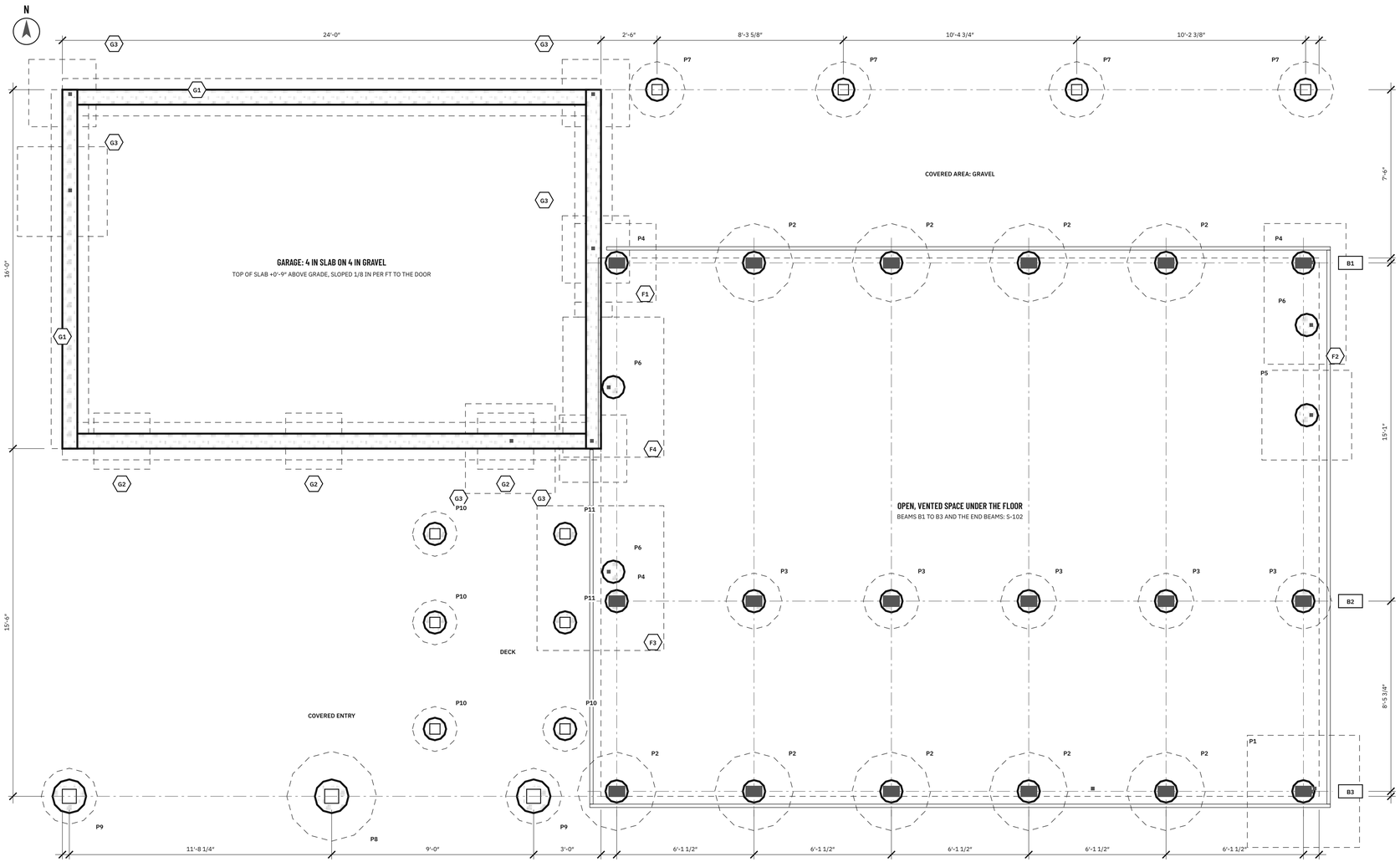
Engineer's stamp

SHEET TITLE

SITE PLAN

SHEET

C-101



1 **FOUNDATION PLAN**

3/8" = 1'-0" Footings and pads dashed below. Pier marks P, shared pads F, garage footings G

FOUNDATION NOTES

1. Planning design from the model for 100 psf roof snow, 115 mph wind and 2,000 psf soil (presumptive, IRC Table R401.4.1). A Colorado-licensed engineer checks it and stamps it; a soils report can change every footing.
2. Concrete 4,000 psi, air-entrained. Rebar ASTM A615 Grade 60. 3 in cover cast against earth, 2 in in forms.
3. Bottom of every footing and pad at least 48 in below finished grade, on undisturbed soil. Call for the footing inspection before placing concrete.
4. Set the beam seats, post bases and hold-down rods to string lines before the concrete sets. Pier tops level within 1/4 in.
5. Hold-down piers (S-401): a threaded rod sized to its hold-down (5/8 to 1 in, F1554 Grade 36) with a heavy nut and plate washer near the bottom of the pier, run up through the end beam to the hold-down. Shared pads F carry two piers, or a pier and the garage wall; pour each pad and its piers as one.
6. Uplift and side load: every beam seat and post base is rated for at least the uplift and side load in the schedule. Do not substitute plain seats.
7. The space under the floor stays open and vented behind the skirt. Water entry and drain drop sit in insulated, heat-traced boxes.
8. Garage: frost footing and stem wall, 4 in slab on 4 in compacted gravel with a vapor retarder, control joints at about 10 ft.
9. Entry deck posts stand on their own 24 in footings, or on a shared pad where one runs under them. The deck is not bolted to the house.
10. Details: A-401 detail 5 (pier) and A-601 (garage).

PIER LOADS FROM THE MODEL

Where	Down	Footing	Soil	Uplift
Beam B1 piers	13,906 lb	42 in round	1,609 psf	1,965 lb
Beam B2 piers	7,564 lb	30 in round	1,742 psf	-
Beam B3 piers	14,172 lb	42 in round	1,636 psf	6,104 lb
Covered-area posts	7,568 lb	30 in round	1,742 psf	-
Entry post at -23'-8 1/4"	6,559 lb	30 in round	1,632 psf	-
Entry post at -12'-0"	16,346 lb	48 in round	1,492 psf	-
Entry post at -3'-0"	8,055 lb	30 in round	1,937 psf	-
Hold-down pier, east wall at 2'-11 7/8"	-	shared pad	-	1,965 lb
Hold-down pier, east wall at 7'-0 1/8"	-	4'-0" square	-	3,781 lb
Hold-down pier, west wall at 5'-9 1/8"	-	shared pad	-	4,927 lb
Hold-down pier, west wall at 13'-11 7/8"	-	shared pad	-	4,927 lb

Down: heaviest of D + L, D + S and D + 0.75(L + S). Soil: pressure under the footing, 2,000 psf allowed. Uplift: wind, with both axes at 75 % where two walls share a pier.

FOUNDATION SCHEDULE

Mark	Qty	What	Pier	Footing	Reinforcing	Cast in
P1	1	House pier under a floor beam	12 in tube, top 1'-6 3/8" above grade	5'-0" square x 12 in pad	4 #4 vertical, hooked; #4 at 12 in each way in the footing, top and bottom	Uplift-rated beam seat: 6,104 lb up, 515 lb sideways
P2	9	House pier under a floor beam	12 in tube, top 1'-6 3/8" above grade	42 in round x 10 in footing, bell form or formed	4 #4 vertical, hooked; #4 at 12 in each way in the footing	Uplift-rated beam seat: 2,329 lb up, 515 lb sideways
P3	5	House pier under a floor beam	12 in tube, top 1'-6 3/8" above grade	30 in round x 10 in footing, bell form or formed	4 #4 vertical, hooked; #4 at 12 in each way in the footing	Uplift-rated beam seat: 500 lb up, 515 lb sideways
P4	3	House pier under a floor beam	12 in tube, top 1'-6 3/8" above grade	On a shared pad (F marks)	4 #4 vertical, hooked into the pad	Uplift-rated beam seat: 1,965 lb up, 515 lb sideways
P5	1	Hold-down pier under an end wall	12 in tube, top 1'-6 3/8" above grade	4'-0" square x 12 in pad	4 #4 vertical, hooked; #4 at 12 in each way in the footing, top and bottom	5/8 in rod for the HDU5 above (S-401)
P6	3	Hold-down pier under an end wall	12 in tube, top 1'-6 3/8" above grade	On a shared pad (F marks)	4 #4 vertical, hooked into the pad	5/8 in rod for the HDU2; 7/8 in rod for the HDU8 above (S-401)
P7	4	Covered-area post pier	12 in tube, top 0'-6" above grade	30 in round x 10 in footing, bell form or formed	4 #4 vertical, hooked; #4 at 12 in each way in the footing	Uplift-rated 6x6 post base: 1,372 lb up
P8	1	Covered-entry post pier	18 in tube, top 0'-6" above grade	48 in round x 10 in footing, bell form or formed	4 #4 vertical, hooked; #4 at 12 in each way in the footing	Uplift-rated 8x8 post base: 1,460 lb up
P9	2	Covered-entry post pier	18 in tube, top 0'-6" above grade	30 in round x 10 in footing, bell form or formed	4 #4 vertical, hooked; #4 at 12 in each way in the footing	Uplift-rated 8x8 post base: 1,460 lb up
P10	4	Entry deck post pier	12 in tube, top 0'-6" above grade	24 in round x 10 in footing, bell form or formed	4 #4 vertical, hooked; #4 at 12 in each way in the footing	6x6 post base
P11	2	Entry deck post pier	12 in tube, top 0'-6" above grade	On a shared pad (F marks)	4 #4 vertical, hooked into the pad	6x6 post base
F1	1	Shared pad: P4 + garage stem wall	-	3'-7 3/8" x 3'-6" x 12 in	#5 at 12 in each way, top and bottom	Pour with the piers it carries
F2	1	Shared pad: P4 + P6	-	3'-7 3/4" x 6'-3 1/4" x 12 in	#5 at 12 in each way, top and bottom	Pour with the piers it carries
F3	1	Shared pad: P4 + P6 + P11 + P11	-	5'-7 3/4" x 6'-5 3/8" x 12 in	#5 at 12 in each way, top and bottom	Pour with the piers it carries
F4	1	Shared pad: P6 + garage stem wall	-	4'-6" x 6'-3" x 12 in	#5 at 12 in each way, top and bottom	Pour with the piers it carries
G1	-	Garage frost footing and stem wall	8 in stem wall to the slab	20 x 10 in continuous	2 #4 continuous in the footing, 2 #4 at the top of the wall, #4 vertical at 48 in	5/8 in anchor bolts under the shear panels as on S-401, 1/2 in at 6 ft elsewhere, 12 in from plate ends
G2	3	Footing widened under a jack post	-	2'-6" square x 10 in	#4 at 8 in each way	Garage south wall
G3	6	Footing thickened under a hold-down	-	4 at 3' square, 6 in deeper; 2 at 4' square, 12 in deeper	#5 at 12 in each way, bottom; stem wall 2 #4 top and bottom	Hold-down anchor cast in (S-401)

RIME CABIN

Owner-built mountain cabin

Summit County, Colorado (Breckenridge area), about 9,500 to 10,000 ft. Lot not chosen yet.

OWNER	Sander Blue
DRAWN BY	Claude (AI)
DATE	2026-10-10
SCALE	3/8" = 1'-0"

OPTIONS SHOWN

Concrete piers · 100 psf roof snow · Propane + wood · Utility power · 3 glass bays · Standing seam · No loft · Washer-dryer · Attached garage

REVISIONS

Rev	Date	Description
A	2026-10-08	Concept model and estimate
B	2026-10-09	Data model, framing, systems, drawings; review fixes
C	2026-10-09	Attached 1-car garage; panel, meter and hood duct moved for it
D	2026-10-10	North roof extended to the garage eave: covered open-air area
E	2026-10-10	Garage roof carried south over the entry; new entry deck
F	2026-10-10	Structural fixes: floor raised 7 in, footings sized from loads, simple-span rafters, garage roof joined to the house, new posts and headers, steel glass-wall posts, wind bracing with hold-down piers

NOT FOR CONSTRUCTION

For planning and permit prep. Structural and code items need a licensed engineer and the building department.

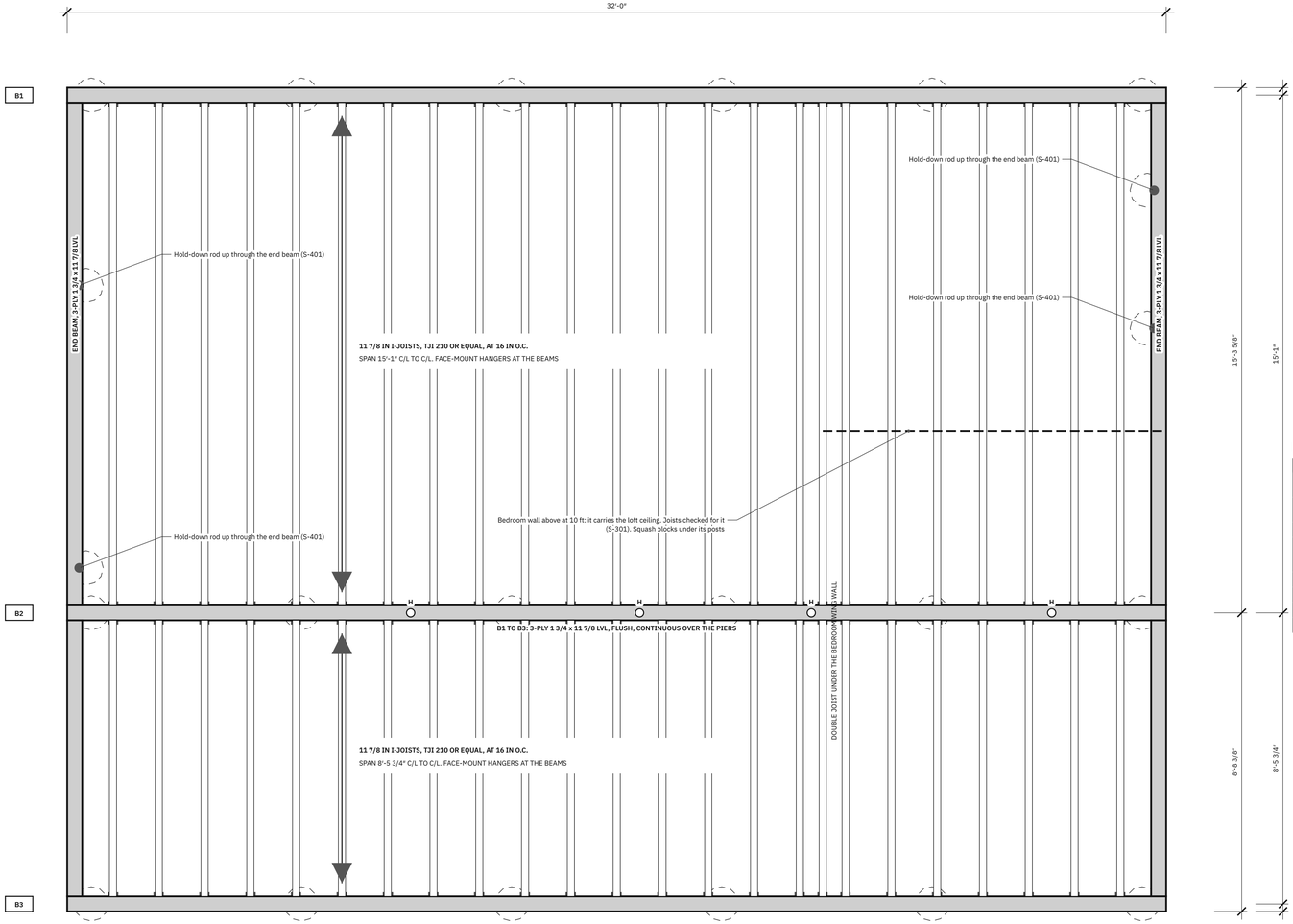
Engineer's stamp

SHEET TITLE

FOUNDATION PLAN

SHEET

S-101



1 **FLOOR FRAMING PLAN**

1/2" = 1'-0" Piers dashed below

FLOOR FRAMING NOTES

1. Joists: 11 7/8 in I-joist, TJI 210 or equal, at 16 in o.c. in face-mount hangers. The series is checked for 40 psf live, 12 psf dead and the bedroom wall and loft load, with L/480 live-load deflection (S-301).
2. Beams: B1 to B3 are 3-ply 1 3/4 x 11 7/8 LVL, flush, continuous over the piers. The east and west walls sit on 3-ply LVL end beams hung between the floor beams in concealed-flange hangers (a rim board cannot span 15 ft under a wall).
3. Subfloor: 3/4 in tongue-and-groove OSB, glued and screwed. Seal all seams; the subfloor is part of the air barrier.
4. Insulation: dense-pack cellulose fills the bays (about R-44). Belly: 1/2 in plywood with rodent mesh under the joists.
5. H: wire holes through beam B2 only where marked, 1 in max, at mid-depth, at least 2 ft from a pier, per the LVL maker. All other wires and pipes go through I-joist webs in the maker's hole zones. Never notch or drill a flange.
6. Hold-down rods (S-401, 5/8 to 1 in) pass straight down through the end beams in holes 1/16 in over the rod size, centered in the beam width, as the LVL maker allows; a pier sits under each one (S-101).
7. Floor diaphragm: the subfloor carries the walls' wind load to the piers: blocked, 10d at 6 in on all edges (S-401). Wall plates screwed down to the beams and rims per S-401.
8. Squash blocks under posts and point loads. Double joist under the bedroom wing wall, which runs along the joists.

FLOOR MATERIALS (FROM THE PARTS LIST)

Item	Spec	Qty
Floor beam, LVL	3-ply 1 3/4 x 11 7/8, flush with the joists. Includes the end beams under the east and west walls (in concealed-flange hangers)	148 lf
Floor joist, I-joist	11 7/8 in at 16 in o.c., TJI 210 or equal (checked on S-301 with the bedroom wall and loft load)	572 lf
I-joist hanger	Face mount, sized to joist	96 ea
End beam hanger, concealed flange	Concealed-flange hanger for 3-ply 1 3/4 x 11 7/8 LVL (HUCQ-type or equal), rated at least 3,100 lb down; hangs the end beams off beams B1 to B3 flush with their ends	8 ea
Subfloor	3/4 in tongue-and-groove, glued and screwed	26 sheet
Floor insulation, R-44	Dense-pack cellulose in 11 7/8 in bays	85 bag
Belly board	1/2 in plywood with rodent mesh under the joists	26 sheet
Subfloor glue, screws, squash blocks		1 lot

RIME CABIN

Owner-built mountain cabin

Summit County, Colorado (Breckenridge area), about 9,500 to 10,000 ft. Lot not chosen yet.

OWNER	Sander Blue
DRAWN BY	Claude (AI)
DATE	2026-10-10
SCALE	1/2" = 1'-0"

OPTIONS SHOWN

Concrete piers · 100 psf roof snow · Propane + wood · Utility power · 3 glass bays · Standing seam · No loft · Washer-dryer · Attached garage

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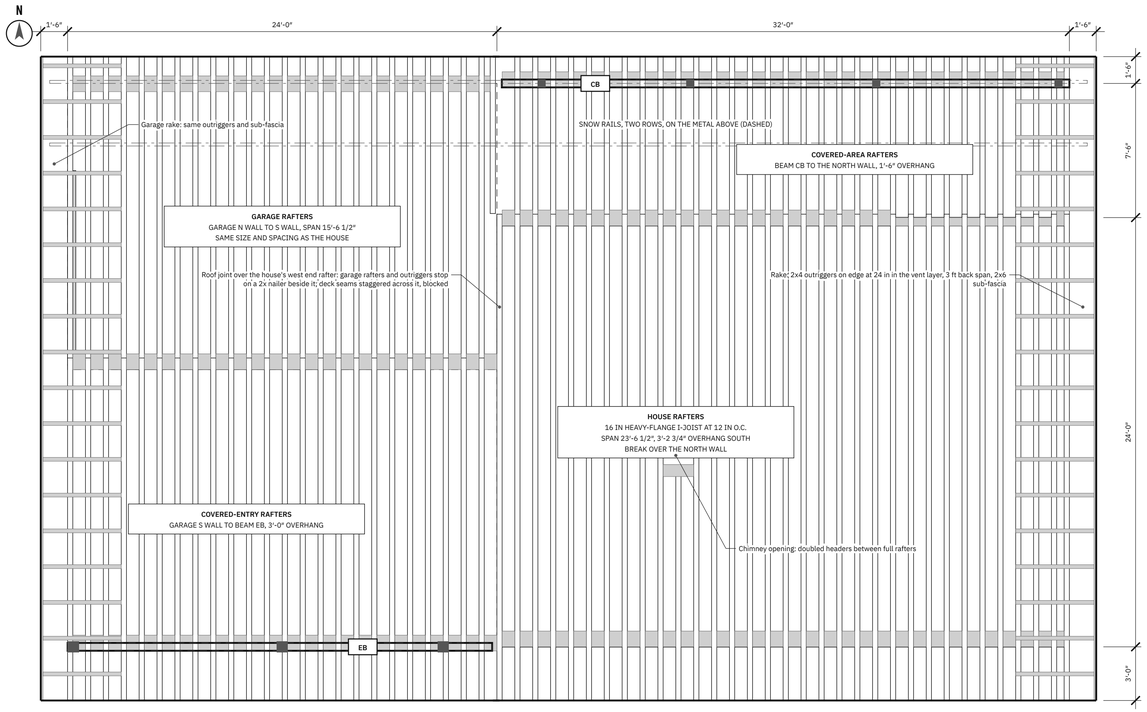
Engineer's stamp

SHEET TITLE

FLOOR FRAMING PLAN

SHEET

S-102

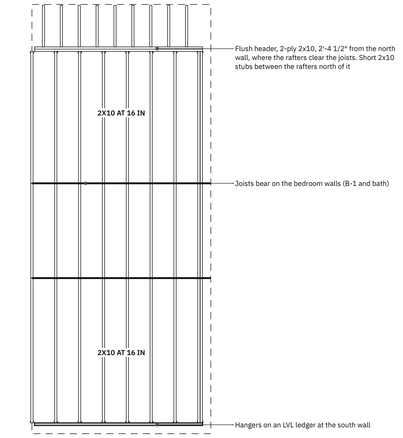


1 ROOF FRAMING PLAN

1/4" = 1'-0" One roof plane over the house, the garage and both covered areas. Decks and metal not shown

ROOF FRAMING NOTES

1. Rafters: 16 in heavy-flange I-joist at 12 in o.c. for 100 psf roof snow (S-301). House rafters run from the north wall to the south eave; the covered-area rafters run from the beam to the north wall and lap the house rafters there with a 2x nailer. Garage rafters break over the garage south wall the same way.
2. Bearing: beveled 2x plates on every wall and beam. No bird's mouth cuts in I-joists. Web stiffeners at every bearing. Blocking panels between rafters at each bearing. A rafter tie (H2.5A type or equal, about 455 lb uplift) at every bearing.
3. Roof deck (the diaphragm): 5/8 in OSB lower deck, blocked, 8d at 4 in on boundaries, 6 in on other edges, 12 in in the field. Then 2x4 sleepers on edge over the rafters (the cold-roof vent channel), then 23/32 in OSB upper deck screwed through the sleepers.
4. Rakes: 2x4 outriggers on edge at 24 in in the vent layer with a 3 ft back span, screwed to the lower deck and the rafters; 2x6 sub-fascia. No ladder framing hung off the end rafter.
5. Garage: the same rafters and roof build-up in the same plane as the house roof. The two meet over the house's west end rafter (see the leader). No step, no cricket, no drift.
6. Snow rails: two rows of clamp-on rail on the metal above the north eave so sliding snow does not land on the covered area.
7. Openings: chimney opening framed with doubled headers between full rafters. Vent pipes pass between rafters.
8. Ceiling joists over the bedroom wing are 2x10 at 16 in for a storage loft (30 psf). Do not notch top or bottom edges.



2 BEDROOM WING CEILING

1/4" = 1'-0"

RAFTER SCHEDULE (SPANS BEARING TO BEARING)

Rafters	Member	Span	Overhang	Reaction	Snow uplift
House	16 in heavy-flange I-joist at 12 in o.c.	23'-6 1/2"	3'-2 3/4"	1,842 lb	none (snow)
Covered area	16 in heavy-flange I-joist at 12 in o.c.	7'-8 3/4"	1'-6" (north)	645 lb	none (snow)
Garage	16 in heavy-flange I-joist at 12 in o.c.	15'-6 1/2"	1'-8 3/4" (north)	1,123 lb	none (snow)
Covered entry	16 in heavy-flange I-joist at 12 in o.c.	15'-8 3/4"	3'-0"	1,305 lb	none (snow)

ROOF BEAMS

Mark	Beam	Member	Supports	Notes
CB	Covered-area beam	3-ply 1 3/4 x 11 7/8 LVL	4 posts, 6x6 DF No. 1, 7'-9 3/8" tall	On a beveled plate under the rafters. Stops short of the garage; the end rafter bears on the garage's short east wall
EB	Covered-entry beam	3-ply 1 3/4 x 16 LVL	3 posts, 8x8 DF No. 1, 17'-9 3/8" tall	On a beveled plate. Stops 1/2 in short of the house siding (flashed); the house does not carry it
GC1	Garage south wall header	3-ply 1 3/4 x 11 7/8 LVL	LVL jack posts	Over the clerestory windows; the garage and entry rafters bear on it
GC2	Garage south wall header	3-ply 1 3/4 x 11 7/8 LVL	LVL jack posts	Over the clerestory windows; the garage and entry rafters bear on it

RIME CABIN

Owner-built mountain cabin

Summit County, Colorado (Breckenridge area), about 9,500 to 10,000 ft. Lot not chosen yet.

OWNER	Sander Blue
DRAWN BY	Claude (AI)
DATE	2026-10-10
SCALE	1/4" = 1'-0"

OPTIONS SHOWN

Concrete piers · 100 psf roof snow · Propane + wood · Utility power · 3 glass bays · Standing seam · No loft · Washer-dryer · Attached garage

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Engineer's stamp

SHEET TITLE

ROOF FRAMING PLANS

SHEET

S-201

DESIGN CRITERIA

Item	Value used	Source and status
Code basis	2021 IRC with engineered parts, ASCE 7-16, NDS 2018, AISC 360, ACI 318	As adopted by the county. Confirm the edition at the permit counter.
Ground and roof snow	100 psf roof snow (option on the cost page)	Summit County publishes the design snow load by elevation. Use the county number for the lot.
Roof dead load	21 psf (house), 17 psf (open areas)	Metal, two OSB decks and sleepers, I-joists, dense-pack, pine ceiling. Open areas have no insulation.
Floor	40 psf live, 12 psf dead	IRC Table R301.5
Loft (storage)	30 psf live, 10 psf dead	IRC Table R301.5
Wind	115 mph, Exposure C, Kd 0.85, Kzt 1.0. qz 24.9 psf at 16′-6″; ASD wall pressure 16.5 psf (MWFRS), 16.2 psf (C&C)	ASCE 7-16 Ch. 27 and 30. 115 mph, Exposure C (Summit County lists 115 mph; C in open areas, B where treed)
Seismic	Category B assumed; wind governs bracing	Confirm with the county
Soil	2,000 psf bearing	2,000 psf presumptive (assumed; get a soils report)
Frost	48 in	48 in used (Summit County lists 40 in; verify for the lot)
Load combinations	D + L, D + S, D + 0.75L + 0.75S, D + 0.6W, D + 0.75(0.6W) + 0.75S, 0.6D + 0.6W	ASCE 7 ASD. Snow duration factor 1.15, wind 1.6
Materials	SPF No. 2, DF-L No. 1 posts, LVL 2.0E, LSL 1.55E, I-joists per maker, steel A500 Gr C	Maker's published values; the engineer confirms

GRAVITY AND MEMBER CHECKS FROM THE MODEL (60 CHECKS, 60 PASS)

Member or check	Size and basis	Demand	Capacity	Used		Ref
House rafters: bending	16 in heavy-flange I-joist at 12 in o.c., span 23′ 7″, 3′ 3″ overhang	8,328 ft-lb	14,864 ft-lb	56%	OK	NDS, maker
House rafters: shear		1,445 lb	3,116 lb	46%	OK	NDS, maker
House rafters: deflection	Total load, L/240, with 15% for I-joist shear deformation	0.73 in	1.18 in	62%	OK	IRC R301.7
House rafters: bearing	Reaction per rafter. Web stiffeners at bearings	1,842 lb	3,910 lb	47%	OK	Maker
Covered-area rafters: bending	16 in heavy-flange I-joist at 12 in o.c., span 7′ 9″	864 ft-lb	14,864 ft-lb	6%	OK	NDS, maker
Covered-area rafters: shear		467 lb	3,116 lb	15%	OK	NDS, maker
Covered-area rafters: deflection	Total load, L/240, with 15% for I-joist shear deformation	0.01 in	0.39 in	2%	OK	IRC R301.7
Covered-area rafters: bearing	Reaction per rafter. Web stiffeners at bearings	645 lb	3,910 lb	16%	OK	Maker
Wall over the bedroom wing: gap under the rafter	The rafter over the wall deflects up to 0.73 in; the wall stops 1 in short, slotted deflection clips at the top, gap sealed with a compressible gasket	0.73 in	1.00 in	73%	OK	Detail
Garage rafters: bending	16 in heavy-flange I-joist at 12 in o.c., span 15′ 7″	3,520 ft-lb	14,864 ft-lb	24%	OK	NDS, maker
Garage rafters: shear		916 lb	3,116 lb	29%	OK	NDS, maker
Garage rafters: deflection	Total load, L/240, with 15% for I-joist shear deformation	0.14 in	0.78 in	18%	OK	IRC R301.7
Garage rafters: bearing	Reaction per rafter. Web stiffeners at bearings	1,123 lb	3,910 lb	29%	OK	Maker
Covered-entry rafters: bending	16 in heavy-flange I-joist at 12 in o.c., span 15′ 9″, 3′ overhang	3,580 ft-lb	14,864 ft-lb	24%	OK	NDS, maker
Covered-entry rafters: shear		949 lb	3,116 lb	30%	OK	NDS, maker
Covered-entry rafters: deflection	Total load, L/240, with 15% for I-joist shear deformation	0.14 in	0.79 in	17%	OK	IRC R301.7
Covered-entry rafters: bearing	Reaction per rafter. Web stiffeners at bearings	1,305 lb	3,910 lb	33%	OK	Maker
End beams under the east and west walls: bending	3-ply 1 3/4 x 11 7/8 LVL, 14′ 8″ span (was a rim board)	12,470 ft-lb	26,775 ft-lb	47%	OK	NDS
End beams: deflection	L/360 under a wall	0.33 in	0.49 in	67%	OK	IRC R301.7
Floor beam B1: bending	3-ply 1 3/4 x 11 7/8 LVL, continuous over piers at 6′ 1″	7,586 ft-lb	26,775 ft-lb	28%	OK	NDS
Beam B1 bearing on the pier caps	Cap seat 5 1/4 x 9 in under the beam, LVL perpendicular-to-grain 750 psi	294 psi	750 psi	39%	OK	Maker
Piers on beam B1: footing	Heaviest pier 13,906 lb on a 42 in round footing	1,609 psf	2,000 psf	80%	OK	IRC R403.1
Floor beam B2: bending	3-ply 1 3/4 x 11 7/8 LVL, continuous over piers at 6′ 1″	4,258 ft-lb	26,775 ft-lb	16%	OK	NDS
Beam B2 bearing on the pier caps	Cap seat 5 1/4 x 9 in under the beam, LVL perpendicular-to-grain 750 psi	160 psi	750 psi	21%	OK	Maker
Piers on beam B2: footing	Heaviest pier 7,564 lb on a 30 in round footing	1,742 psf	2,000 psf	87%	OK	IRC R403.1
Floor beam B3: bending	3-ply 1 3/4 x 11 7/8 LVL, continuous over piers at 6′ 1″	7,790 ft-lb	26,775 ft-lb	29%	OK	NDS
Beam B3 bearing on the pier caps	Cap seat 5 1/4 x 9 in under the beam, LVL perpendicular-to-grain 750 psi	300 psi	750 psi	40%	OK	Maker
Piers on beam B3: footing	Heaviest pier 14,172 lb on a 42 in round footing	1,636 psf	2,000 psf	82%	OK	IRC R403.1
Ceiling joists over the bedroom wing (loft floor)	2x10 SPF No. 2 at 16 in o.c., longest span 8′ 1″, 30 psf storage + 10 psf dead; live deflection 0.03 in (L/3480)	437 ft-lb	1,973 ft-lb	22%	OK	NDS, IRC R301.5
Ceiling header near the north wall	2-ply 2x10 SPF No. 2 in hangers, 9′ 5″ span, 200 plf; live deflection 0.09 in (L/1190)	2,204 ft-lb	3,432 ft-lb	64%	OK	NDS

Member or check	Size and basis	Demand	Capacity	Used		Ref
Floor joists: bending	11 7/8 in I-joist, TJI 210 or equal, at 16 in o.c., clear span 14′ 8″, 40 psf live, 12 psf dead, plus the bedroom partition and loft at 10′	3,155 ft-lb	3,795 ft-lb	83%	OK	Maker, IRC R301.5
Floor joists: shear		795 lb	1,655 lb	48%	OK	Maker
Floor joists: live-load deflection	L/480 used (IRC asks L/360); stiffer floor for tile and plank	0.28 in	0.37 in	77%	OK	IRC R301.7
Covered-area beam: bending	3-ply 1 3/4 x 11 7/8 LVL on posts at 2′ 6″, 10′ 10″, 21′ 2″, 31′ 5″	7,231 ft-lb	30,791 ft-lb	23%	OK	NDS
Covered-area posts	6x6 Douglas fir-larch No. 1, 7′ 9″ tall	7,568 lb	27,812 lb	27%	OK	NDS 3.7
Covered-area post footings	30 in round bell under a 12 in pier	1,742 psf	2,000 psf	87%	OK	IRC R403.1
Covered-entry beam: bending	3-ply 1 3/4 x 16 LVL on 8x8 posts at -23′ 8″, -12′, -3′; it stops short of the house	17,114 ft-lb	53,692 ft-lb	32%	OK	NDS
Covered-entry posts	8x8 Douglas fir-larch No. 1, 17′ 9″ tall	16,346 lb	28,598 lb	57%	OK	NDS 3.7
Covered-entry post footings	Worst: the post at -3′, 30 in round footing under an 18 in pier	1,937 psf	2,000 psf	97%	OK	IRC R403.1
Garage south wall footing	20 in continuous footing, 8 in stem wall; pads 2 ft 6 in square under the jack posts	1,477 psf	2,000 psf	74%	OK	IRC R403.1
Header N1 (N wall)	2-ply 1 3/4 x 11 7/8 LVL, 2′ 9″ span, 1969 plf. 1 jack stud each end (end reaction 2,707 lb)	1,861 ft-lb	20,528 ft-lb	9%	OK	NDS
Header N2 (N wall)	2-ply 1 3/4 x 11 7/8 LVL, 3′ 3″ span, 1969 plf. 1 jack stud each end (end reaction 3,199 lb)	2,599 ft-lb	20,528 ft-lb	13%	OK	NDS
Header S4 (S wall)	2-ply 1 3/4 x 11 7/8 LVL, 3′ 3″ span, 2029 plf. 1 jack stud each end (end reaction 3,297 lb)	2,679 ft-lb	20,528 ft-lb	13%	OK	NDS
Header E1 (E wall)	2-ply 2x10 SPF No. 2, 3′ 3″ span, 430 plf. 1 jack stud each end (end reaction 699 lb)	568 ft-lb	3,947 ft-lb	14%	OK	NDS
Header W1 (W wall)	2-ply 2x10 SPF No. 2, 3′ 3″ span, 264 plf. 1 jack stud each end (end reaction 429 lb)	349 ft-lb	3,947 ft-lb	9%	OK	NDS
Header W3 (W wall)	2-ply 2x10 SPF No. 2, 3′ 5″ span, 264 plf. 1 jack stud each end (end reaction 455 lb)	393 ft-lb	3,947 ft-lb	10%	OK	NDS
Header W4 (W wall)	2-ply 2x10 SPF No. 2, 3′ 3″ span, 264 plf. 1 jack stud each end (end reaction 429 lb)	349 ft-lb	3,947 ft-lb	9%	OK	NDS
Header GH (W wall)	2-ply 2x10 SPF No. 2, 3′ 3″ span, 264 plf. 1 jack stud each end (end reaction 429 lb)	349 ft-lb	3,947 ft-lb	9%	OK	NDS
Header SB1 (S wall)	2-ply 1 3/4 x 11 7/8 LVL, 6′ 8″ span, 2029 plf. 2 jack studs each end (end reaction 6,746 lb)	11,215 ft-lb	20,528 ft-lb	55%	OK	NDS
Header SB2 (S wall)	2-ply 1 3/4 x 11 7/8 LVL, 6′ 8″ span, 2029 plf. 2 jack studs each end (end reaction 6,746 lb)	11,215 ft-lb	20,528 ft-lb	55%	OK	NDS
Header SB3 (S wall)	2-ply 1 3/4 x 11 7/8 LVL, 6′ 8″ span, 2029 plf. 2 jack studs each end (end reaction 6,746 lb)	11,215 ft-lb	20,528 ft-lb	55%	OK	NDS
Posts between the glass bays: snow plus wind	HSS 5 x 3 x 1/4 steel tube (A500 Gr C), 17′ 1″ tall, 13,492 lb roof load, wind 16 psf on 7′ of wall; transoms brace it in plane every 7′ or less. (A 4-ply LSL post would be at 110%)	0.49 (ratio)	1.00	49%	OK	AISC 360 E3, H1-1; ASCE 7 C&C
Posts between the glass bays: wind deflection	L/175 for the glass framing at 0.42 x the C&C wind	0.48 in	1.17 in	41%	OK	IRC Table R301.7
Tall wall studs (LSL): snow plus wind	1 3/4 x 5 1/2 LSL (1.55E) at 16 in o.c. in the south wall, 17′ 1″ tall, 2,705 lb roof load each, wind 16 psf	0.66 (ratio)	1.00	66%	OK	NDS 3.9.2, ASCE 7 C&C
Tall wall studs (LSL): wind deflection	L/180 at 0.42 x the C&C wind	0.76 in	1.14 in	67%	OK	IRC Table R301.7
Covered-area post footings: wind uplift	About 21 psf net on the open roof; 1,372 lb at the worst post, held by its footing and the soil over it. Post base and beam cap rated at least this much	1,372 lb	1,375 lb	100%	OK	ASCE 7 (0.6D + 0.6W)
Covered-entry post footings: wind uplift	About 21 psf net on the open roof; 1,460 lb at the post at -3′, held by its footing and the soil over it	1,460 lb	1,468 lb	99%	OK	ASCE 7 (0.6D + 0.6W)
Rafter ties: wind uplift	Net uplift at the eave bearing about 222 lb per rafter (27 psf suction less 0.6 x dead load); tie rated about 455 lb (H2.5A-type, SPF)	222 lb	455 lb	49%	OK	ASCE 7 C&C, IRC R802.11
Garage header GC1	3-ply 1 3/4 x 11 7/8 LVL, 8′ 3″ span, 1855 plf, deflection 0.13 in	15,781 ft-lb	30,791 ft-lb	51%	OK	NDS
Garage header GC2	3-ply 1 3/4 x 11 7/8 LVL, 8′ 3″ span, 1855 plf, deflection 0.13 in	15,781 ft-lb	30,791 ft-lb	51%	OK	NDS

CONNECTION SCHEDULE (COUNTS FROM THE MODEL)

Connector or part	Where and what	Qty
Beam-to-pier cap, uplift rated	Cast-in steel beam seat (fabricated U-seat of 3/16 in plate with welded anchor straps, galvanized), 5 1/4 x 9 in bearing, through-bolted to the beam; rated for the uplift and side load on S-101	18 ea
Hold-down	Sized on S-401 (HDU2 to HDU8 type). House: 5/8 in threaded rod cast into a pier (S-101) and run up through the floor framing. Garage: anchor cast into the footing	14 ea
I-joist hanger	Face mount, sized to joist	96 ea
End beam hanger, concealed flange	Concealed-flange hanger for 3-ply 1 3/4 x 11 7/8 LVL (HUCQ-type or equal), rated at least 3,100 lb down; hangs the end beams off beams B1 to B3 flush with their ends	8 ea
Post between glass bays, steel	HSS 5 x 3 x 1/4 steel tube (ASTM A500 Gr C), full height, 1/2 in base and cap plates and seat angles for the bay headers welded on, shop primed. A 2x wood buck lag-bolted on each side for the windows. Sized on S-301 (a 4-ply LSL post is too weak here)	33 lf
Header seat at the steel posts	Steel seat angle welded to the post at each bay header end (on the post shop drawing); the LVL header sits on it and is through-bolted. The wood bucks stop under the header and start again above it	4 ea
Interior shear panel	15/32 in rated sheathing on the inside face of the marked shear walls (S-401), 8d nails at 4 or 3 in on the edges, blocked. Drywall goes over it	20 sheet
Wall-to-floor strap	Coiled strap at corners and beside big openings	8 ea
Rafter tie	Rafter-to-plate tie rated about 455 lb uplift in SPF at every bearing (H2.5A-type for I-joists)	98 ea
Joist hanger, 2x10	At the south ledger	20 ea
Rafter tie	Rafter-to-plate tie rated about 455 lb uplift in SPF at every bearing	72 ea

STRUCTURAL NOTES

1. Every size here comes from the model's own load path, roof to footing, for the options shown. They are planning checks (ASD, simple spans and continuous beams, NDS and maker values), not an engineered design. A Colorado-licensed engineer checks them and stamps the drawings.
2. Load path: roof snow into the rafters, then the north and south walls, the covered-area beam and posts, the garage walls and the covered-entry beam and posts, then the floor beams and end beams, then the piers and footings.
3. South glass wall: the posts between the bays are steel tubes; a 4-ply LSL post fails there once wind on the glass is added to the roof load. The LSL studs are checked the same way.
4. Wind bracing, hold-downs, end posts, base fastening, the roof and floor diaphragms and the piers under wind: S-401.
5. Snow: single shed roof with no higher roof beside it, so no drift. Sliding snow lands north of the eave; the snow rails keep it off the covered area.

RIME CABIN

Owner-built mountain cabin

Summit County, Colorado (Breckenridge area), about 9,500 to 10,000 ft. Lot not chosen yet.

OWNER	Sander Blue
DRAWN BY	Claude (AI)
DATE	2026-10-10
SCALE	None

OPTIONS SHOWN

Concrete piers · 100 psf roof snow · Propane + wood · Utility power · 3 glass bays · Standing seam · No loft · Washer-dryer · Attached garage

REVISIONS

Rev	Date	Description
A	2026-10-08	Concept model and estimate
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NOT FOR CONSTRUCTION

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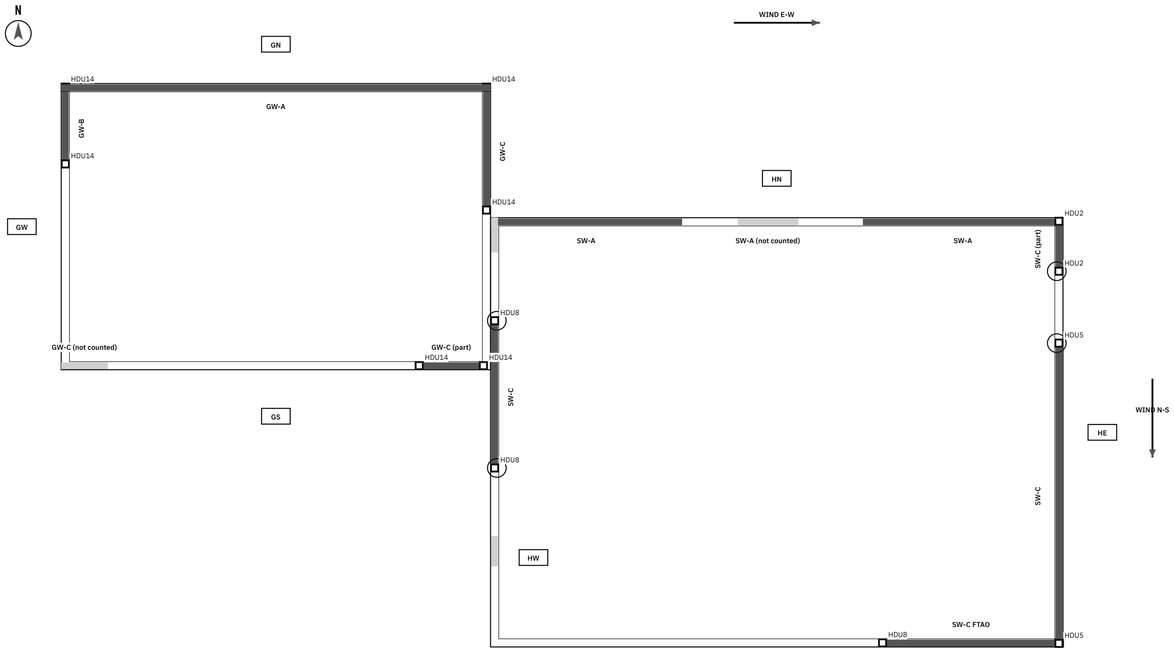
Engineer's stamp

SHEET TITLE

DESIGN CRITERIA AND MEMBER CHECKS

SHEET

S-301



1 **WIND BRACING PLAN**

1/4" = 1'-0" Black bars: shear wall panels. Squares: hold-downs. Circles: hold-down piers (S-101)

WIND BRACING NOTES

1. Method: ASCE 7-16 directional procedure. The roof deck is a flexible diaphragm that hands each strip of wind to the wall lines on either side. Each line's shear walls are its full-height solid panels, sharing the load by strength. Wood-panel walls taller than 2:1 count at 2L/h up to 3.5:1; ZIP-R walls count only up to 2:1 (SDPWS 4.3, ESR-3373).
2. Capacities are wind ASD values for SPF framing (x 0.92). ZIP System R-6 needs 0.131 x 3 in nails so they reach 1 1/2 in into the studs through the foam. Where an inside panel is added, the two faces combine as the larger of twice the weaker or the stronger (SDPWS 4.3.3.2).
3. Hold-downs: HDU type or equal on the end post listed, SPF values. Rod sizes follow the hold-down. House: threaded rod down through the floor framing into a pier (S-101). Garage: anchor into the footing; a stem wall with 2 #4 top and bottom lifts the footing on both sides.
4. Uplift where two walls' hold-downs share a pier or pad: wind on both axes at 75 % at once (ASCE 7 Case 3) is checked as well as each axis alone.
5. Roof diaphragm: 5/8 in OSB lower deck, blocked, 8d at 4 in on boundaries, 6 in on other edges, 12 in in the field. Unit shear 402 plf of 505 plf allowed. Floor diaphragm: 3/4 in OSB subfloor, glued, blocked, 10d at 6 in on all edges; 271 plf of 274 allowed.
6. Collectors: where a line's panels are shorter than the wall, the double top plate carries the rest to them; lap the plates 4 ft and add the strap listed over each top plate splice.
7. Piers: about 515 lb per pier sideways (22 piers share the house's wind). Bending 1,819 ft-lb of about 9,000 allowed; soil and footing checked on S-301.
8. Interior shear panels (SW-C, SW-D): 15/32 in rated sheathing on the inside face, nailed as listed, blocked; drywall goes over it. FTAO straps, if any, are detailed by the engineer.

WIND LOADS

Wind basis	Value
Basic wind speed, exposure	115 mph, Exposure C (ASCE 7-16)
Mean roof height	16′-6″
Kz, qz	0.87, 24.9 psf
Design pressure (ASD, walls + roof)	16.5 psf on the projected area
Total, N-S wind	20,199 lb
Total, E-W wind	9,482 lb

SHEAR WALL LINES

Line	Wall	Wind	Load	Type	Panels (effective)	Max shear	Used
HN	House north wall	E-W	2,225 lb	SW-A	2 (21′-10 1/2″)	102 plf	34%
HS	House south wall	E-W	3,576 lb	SW-C FTAO	1 (10′-3 1/2″)	348 plf	59%
HE	House east wall	N-S	6,507 lb	SW-C	2 (19′-3 3/4″)	337 plf	57%
HW	House west wall and the garage east wall	N-S	9,648 lb	SW-C + GW-C	2 (15′-10 3/4″)	812 plf	73%
GW	Garage west wall	N-S	4,044 lb	GW-B	1 (4′-8 1/4″)	862 plf	100%
GN	Garage north wall	E-W	1,527 lb	GW-A	1 (24′-0″)	64 plf	21%
GS	Garage south wall	E-W	2,155 lb	GW-C	1 (2′-3 7/8″)	540 plf	84%

HOLD-DOWNS, ONE AT EACH END OF EACH PANEL (SPF VALUES)

Line	Panel	Height	Uplift	Hold-down	Rod	End post	Anchored
HS	21′-8 1/2″ to 32′-0″	17′-0 1/2″	4,358 lb	HDU8	7/8 in	2-ply LSL	Rod through beam B3 to its piers
HE	0′-0″ to 3′-2 1/4″	9′-6 7/8″	1,965 lb	HDU2	5/8 in	2-ply 2x6	Rod into a pier (S-101)
HE	6′-9 3/4″ to 24′-0″	14′-2 1/8″	3,781 lb	HDU5	5/8 in	2-ply LSL	Rod into a pier (S-101)
HW	5′-6 3/4″ to 14′-2 1/4″	12′-4″	4,927 lb	HDU8	7/8 in	2-ply LSL	Rod into a pier (S-101)
HW	-7′-6″ to -0′-2 3/4″	9′-7″	7,426 lb	HDU14	1 in	4-ply 2x6	Garage footing, pad G3
GW	-7′-6″ to -2′-9 3/4″	9′-1 7/8″	7,727 lb	HDU14	1 in	4-ply 2x6	Garage footing, pad G3
GS	-4′-2 1/4″ to -0′-2 3/8″	13′-8 1/2″	7,054 lb	HDU14	1 in	4-ply LSL	Garage footing, pad G3

BASE FASTENING AND COLLECTORS

Line	Base of the panels (shear into the floor or footing)	Collector (top plate strap)
HN	1/4 x 6 in structural screws at 16 in into the beam or rim	469 lb, CS16
HS	1/4 x 6 in structural screws at 10 in into the beam or rim	2,426 lb, CMST14
HE	1/4 x 6 in structural screws at 10 in into the beam or rim	981 lb, CS16
HW	1/4 x 6 in structural screws at 8 in into the beam or rim; 5/8 in anchor bolts at 12 in, 3 x 3 x 0.229 in plate washers	3,625 lb, CMST14
GW	5/8 in anchor bolts at 12 in, 3 x 3 x 0.229 in plate washers	2,859 lb, CMST14
GN	5/8 in anchor bolts at 48 in, 3 x 3 x 0.229 in plate washers	-
GS	5/8 in anchor bolts at 18 in, 3 x 3 x 0.229 in plate washers	1,559 lb, CS16

SHEAR WALL TYPES (WIND, ASD, SPF FRAMING)

Type	Sheathing and nailing	Allowable
GW-A	7/16 ZIP sheathing outside, 8d nails at 6 in on the edges, 12 in in the field, blocked	297 plf
GW-B	7/16 ZIP outside and the 7/16 OSB lining inside, both 8d at 4 in on the edges, blocked	865 plf
GW-C	7/16 ZIP outside and the OSB lining inside, both 8d at 3 in on the edges, blocked	1109 plf
SW-A	ZIP System R-6, 0.131 x 3 in nails at 4 in on the panel edges, 12 in in the field, blocked (2:1 max, ESR-3373)	296 plf
SW-C	SW-A outside plus 15/32 rated sheathing on the inside face, 8d at 4 in on the edges, blocked (drywall over it)	592 plf

WIND CHECKS FROM THE MODEL (42, 42 PASS)

Check	Demand	Capacity	Used	
End post HN at 0′ to 10′ 8″	2,793 lb	7,013 lb	40%	OK
End post HN at 13′ 10″ to 17′ 2″	2,103 lb	7,013 lb	30%	OK
End post HN at 20′ 10″ to 32′	2,793 lb	7,013 lb	40%	OK
Wall base shear, House north wall	136 lb	330 lb	41%	OK
Wind bracing, House north wall	2,225 lb	6,482 lb	34%	OK
Hold-down HS at 21′ 9″ to 32′	4,358 lb	5,820 lb	75%	OK
End post HS at 21′ 9″ to 32′	6,639 lb	7,013 lb	95%	OK
Wall base shear, House south wall	290 lb	330 lb	88%	OK
Wind bracing, House south wall	3,576 lb	6,097 lb	59%	OK
Hold-down HE at 0′ to 3′ 2″	1,965 lb	2,215 lb	89%	OK
End post HE at 0′ to 3′ 2″	2,406 lb	7,013 lb	34%	OK
Hold-down HE at 6′ 10″ to 24′	3,781 lb	4,340 lb	87%	OK
End post HE at 6′ 10″ to 24′	5,033 lb	7,013 lb	72%	OK
Wall base shear, House east wall	281 lb	330 lb	85%	OK
Wind bracing, House east wall	6,507 lb	11,444 lb	57%	OK
End post HW at 0′ to 1′ 11″	319 lb	7,013 lb	5%	OK
Hold-down HW at 5′ 7″ to 14′ 2″	4,927 lb	5,820 lb	85%	OK
End post HW at 5′ 7″ to 14′ 2″	5,571 lb	7,013 lb	79%	OK
End post HW at 17′ 10″ to 19′ 6″	319 lb	7,013 lb	5%	OK
Hold-down HW at -7′ 6″ to -0′ 3″	7,426 lb	10,350 lb	72%	OK
End post HW at -7′ 6″ to -0′ 3″	8,003 lb	14,025 lb	57%	OK
Wall base shear, House west wall and the garage east wall (on the floor)	289 lb	330 lb	88%	OK
Wall base shear, House west wall and the garage east wall (on concrete)	812 lb	896 lb	91%	OK
Wind bracing, House west wall and the garage east wall	9,648 lb	13,173 lb	73%	OK
Hold-down GW at -7′ 6″ to -2′ 10″	7,727 lb	10,350 lb	75%	OK
End post GW at -7′ 6″ to -2′ 10″	8,056 lb	14,025 lb	57%	OK
Wall base shear, Garage west wall	862 lb	896 lb	96%	OK
Wind bracing, Garage west wall	4,044 lb	4,056 lb	100%	OK
End post GN at -24′ to 0′	1,672 lb	7,013 lb	24%	OK
Wall base shear, Garage north wall	254 lb	896 lb	28%	OK
Wind bracing, Garage north wall	1,527 lb	7,121 lb	21%	OK
End post GS at -24′ to -21′ 5″	1,951 lb	7,013 lb	28%	OK
Hold-down GS at -4′ 2″ to -0′ 2″	7,054 lb	10,350 lb	68%	OK
End post GS at -4′ 2″ to -0′ 2″	7,791 lb	14,025 lb	56%	OK
Wall base shear, Garage south wall	810 lb	896 lb	90%	OK
Wind bracing, Garage south wall	2,155 lb	2,575 lb	84%	OK
Hold-downs into the garage footing	7,727 lb	8,338 lb	93%	OK
Hold-down uplift on the floor-beam piers	6,104 lb	6,821 lb	89%	OK
Roof diaphragm (lower deck)	402 plf	505 plf	80%	OK
Floor diaphragm (subfloor)	271 plf	274 plf	99%	OK
Piers in bending under wind	1,819 ft-lb	9,000 ft-lb	20%	OK
Piers: lateral load on the soil	515 lb	646 lb	80%	OK

RIME CABIN

Owner-built mountain cabin

Summit County, Colorado (Breckenridge area), about 9,500 to 10,000 ft. Lot not chosen yet.

OWNER	Sander Blue
DRAWN BY	Claude (AI)
DATE	2026-10-10
SCALE	1/4" = 1'-0"

OPTIONS SHOWN

Concrete piers · 100 psf roof snow · Propane + wood · Utility power · 3 glass bays · Standing seam · No loft · Washer-dryer · Attached garage

REVISIONS

Rev	Date	Description
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NOT FOR CONSTRUCTION

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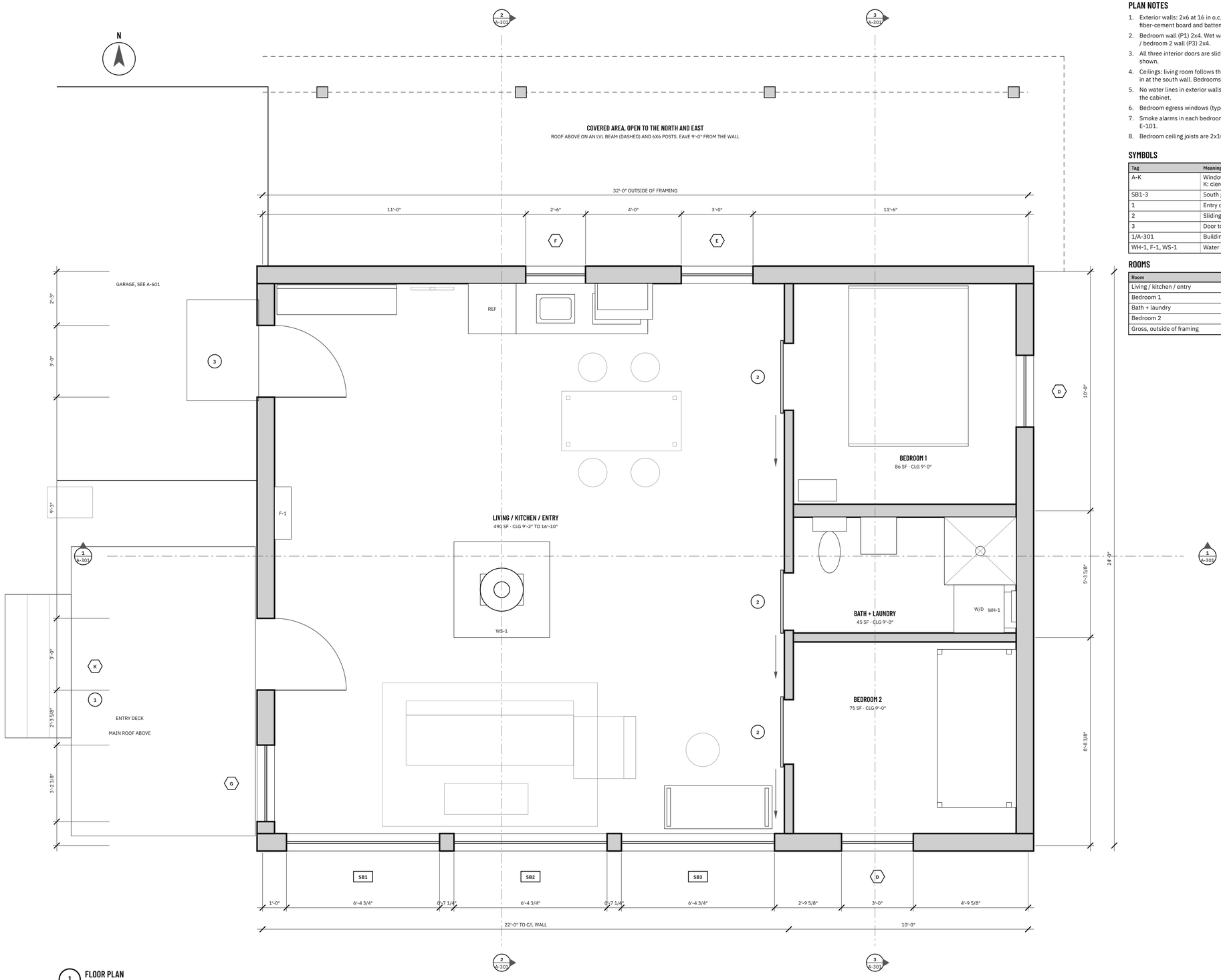
Engineer's stamp

SHEET TITLE

WIND BRACING PLAN

SHEET

S-401



1 **FLOOR PLAN**

1/2" = 1'-0" Dimensions to outside of framing and center of partitions

PLAN NOTES

1. Exterior walls: 2x6 at 16 in o.c., R-23 batts, ZIP System R-6 sheathing taped, rainscreen, fiber-cement board and batten. Drywall inside.
2. Bedroom wall (P1) 2x4. Wet wall between bedroom 1 and bath (P2) 2x6 for the 3 in stack. Bath / bedroom 2 wall (P3) 2x4.
3. All three interior doors are sliding barn doors hung on the living room side. Slide direction shown.
4. Ceilings: living room follows the roof, pine boards, 9 ft-2 in at the north wall to 16 ft-10 in at the south wall. Bedrooms and bath 9 ft-0 in drywall.
5. No water lines in exterior walls. Kitchen supply and drain come up through the floor inside the cabinet.
6. Bedroom egress windows (type D): 36 x 48 in casement, sill 36 in above the floor.
7. Smoke alarms in each bedroom, smoke/CO alarms in the living room near each bedroom door. See E-101.
8. Bedroom ceiling joists are 2x10 so a storage loft can be added later.

SYMBOLS

Tag	Meaning
A-K	Window type, see the schedule on A-501. G with J above it: tall corner window. K: clerestory over the door
SB1-3	South glass bay: two fixed units over a high transom or awning
1	Entry door, 36 in insulated fiberglass
2	Sliding barn door
3	Door to the garage, solid-core steel, self-closing. See A-601
1/A-301	Building section mark
WH-1, F-1, WS-1	Water heater, wall furnace, wood stove. See M-101

ROOMS

Room	Size	Area	Ceiling
Living / kitchen / entry	21'-3 3/4" x 23'-0"	490 SF	Sloped
Bedroom 1	9'-3 3/4" x 9'-2 3/4"	86 SF	9'-0"
Bath + laundry	9'-3 3/4" x 4'-10 1/8"	45 SF	9'-0"
Bedroom 2	9'-3 3/4" x 8'-0 1/8"	75 SF	9'-0"
Gross, outside of framing	32'-0" x 24'-0"	768 SF	

RIME CABIN

Owner-built mountain cabin

Summit County, Colorado (Breckenridge area), about 9,500 to 10,000 ft. Lot not chosen yet.

OWNER	Sander Blue
DRAWN BY	Claude (AI)
DATE	2026-10-10
SCALE	1/2" = 1'-0"

OPTIONS SHOWN

Concrete piers · 100 psf roof snow · Propane + wood · Utility power · 3 glass bays · Standing seam · No loft · Washer-dryer · Attached garage

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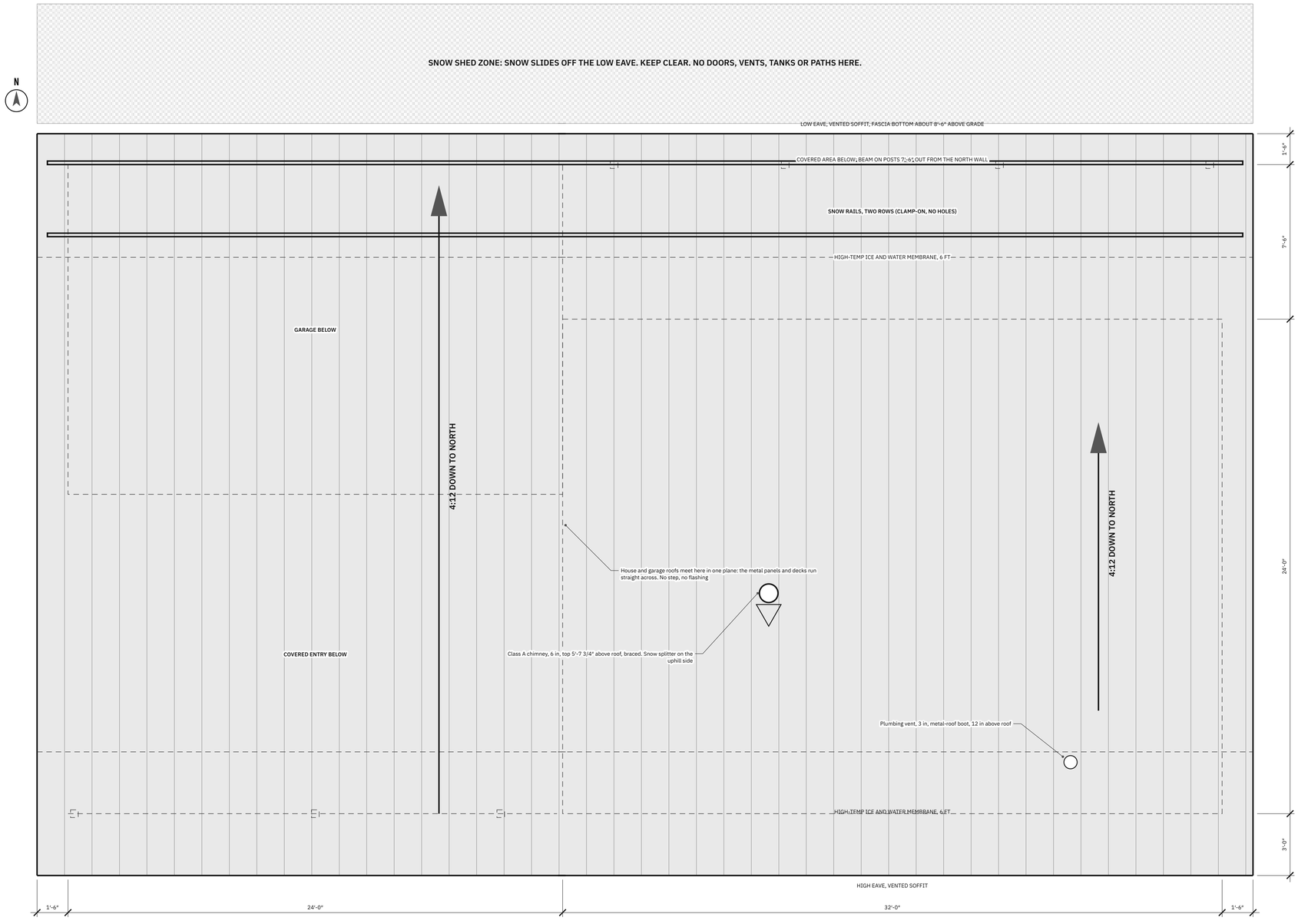
Engineer's stamp

SHEET TITLE

FLOOR PLAN

SHEET

A-101



1 **ROOF PLAN**

3/8" = 1'-0" The house, the garage and the covered areas under one shed roof

ROOF NOTES

1. Roof: 24 ga standing seam steel, snap-lock, concealed clips, panels run north-south in one length (about 38 ft), no end laps.
2. Underlayment: high-temp ice and water membrane 6 ft up from both eaves, synthetic underlayment elsewhere.
3. Cold roof: 2x4 sleepers on edge over the rafters make a 3 1/2 in air channel from the north soffit to the south soffit. Vent both soffits full length. Upper deck 23/32 in OSB.
4. No gutters. Snow and rain leave the low north eave. Keep the snow shed zone clear.
5. Snow rails: two rows of clamp-on rail above the north eave hold sliding snow back so it does not land on the covered area or the garage doors. Clamp to the seams; no holes in the metal.
6. Keep penetrations near the high (south) edge where possible. The plumbing vent rises at the high side for this reason. The chimney gets a snow splitter.
7. Rakes: 1 ft-6 in overhang on 2x4 outriggers in the vent layer, 2x6 sub-fascia. Metal rake trim.
8. Garage: the house roof carried west in the same plane, same build-up, one run of panels. It also runs south over the covered entry to the house's south eave line.
9. Covered area: the roof runs on past the north wall to an LVL beam on posts, flush with the garage eave. Cedar ceiling below; the vent channel runs through to the eave soffit.
10. Framing under this roof: S-201. Lightning, solar and antenna mounts are not shown. Do not penetrate the roof without flashing details.

RIME CABIN

Owner-built mountain cabin

Summit County, Colorado (Breckenridge area), about 9,500 to 10,000 ft. Lot not chosen yet.

OWNER	Sander Blue
DRAWN BY	Claude (AI)
DATE	2026-10-10
SCALE	3/8" = 1'-0"

OPTIONS SHOWN

Concrete piers · 100 psf roof snow · Propane + wood · Utility power · 3 glass bays · Standing seam · No loft · Washer-dryer · Attached garage

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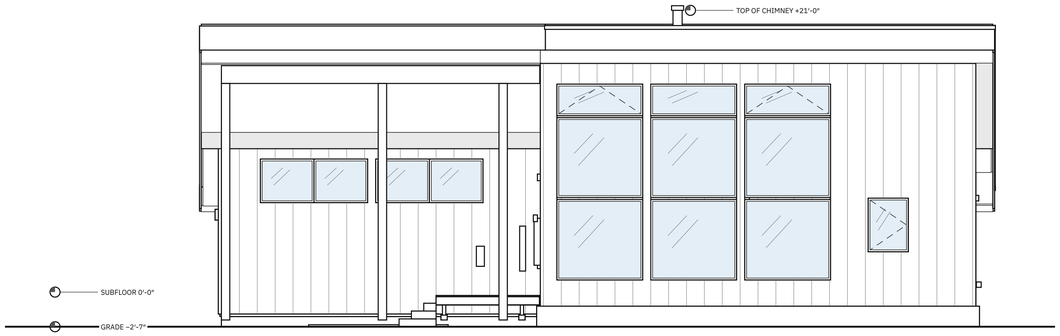
Engineer's stamp

SHEET TITLE

ROOF PLAN

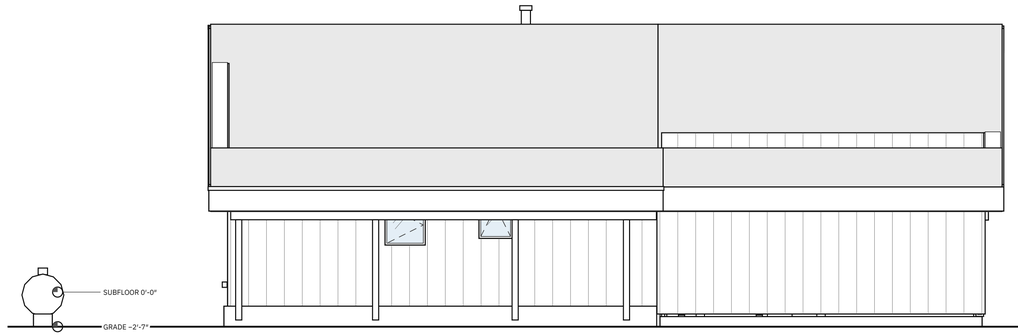
SHEET

A-102



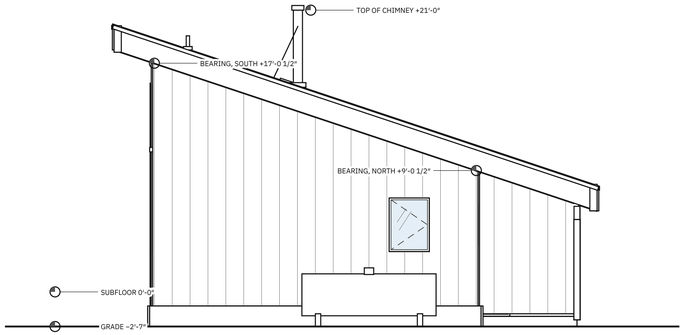
1 SOUTH ELEVATION

3/16" = 1'-0" With the attached garage



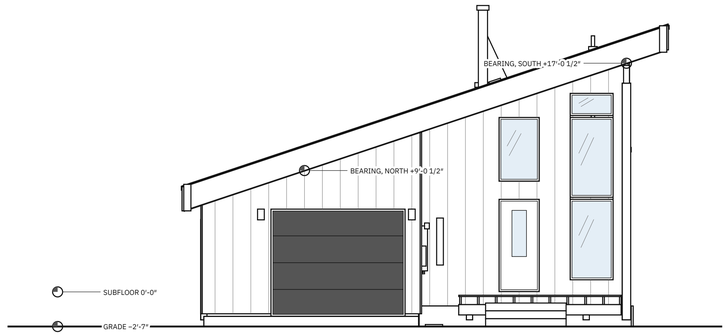
2 NORTH ELEVATION

3/16" = 1'-0" With the attached garage



3 EAST ELEVATION

3/16" = 1'-0"



4 WEST ELEVATION

3/16" = 1'-0" With the attached garage

1. Siding: fiber-cement board and batten, battens at 16 in o.c., over a rainscreen mat on ZIP System R-6 sheathing. Paint. Trim and corners fiber-cement.
2. Roof: standing seam steel, dark. Fascia and trim metal-wrapped. Soffits cedar, vented. Windows: dark frames, low-E, U-0.30 or better.
3. Grade shown level for clarity. Actual grades from the site survey. Keep 8 in from grade to siding. Levels are measured from the top of the subfloor.

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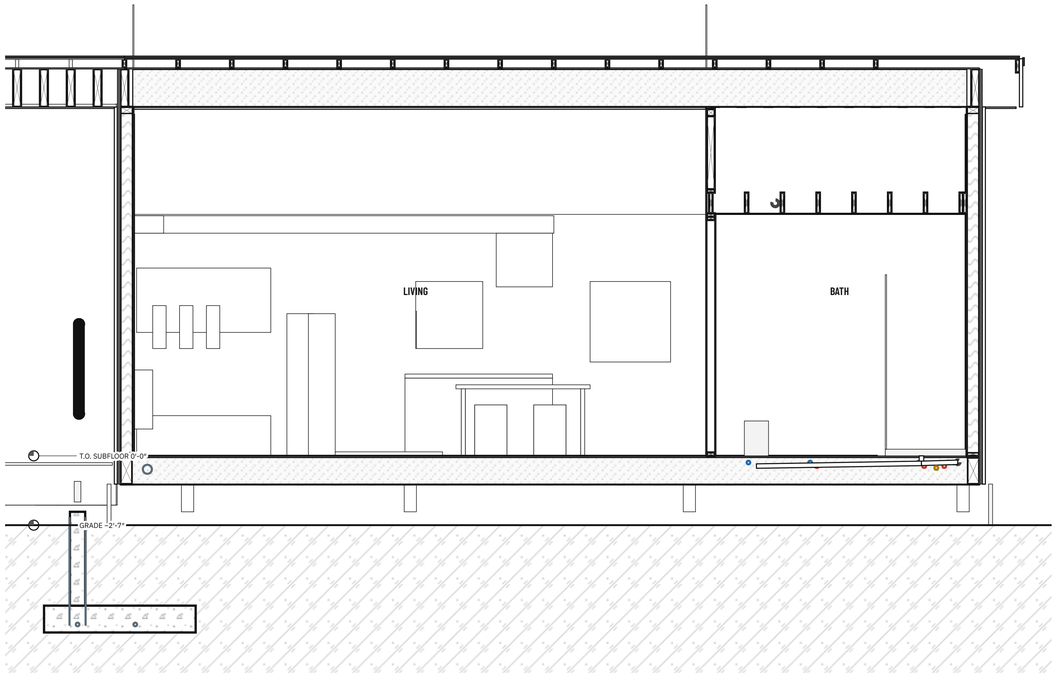
Engineer's stamp

SHEET TITLE

EXTERIOR ELEVATIONS

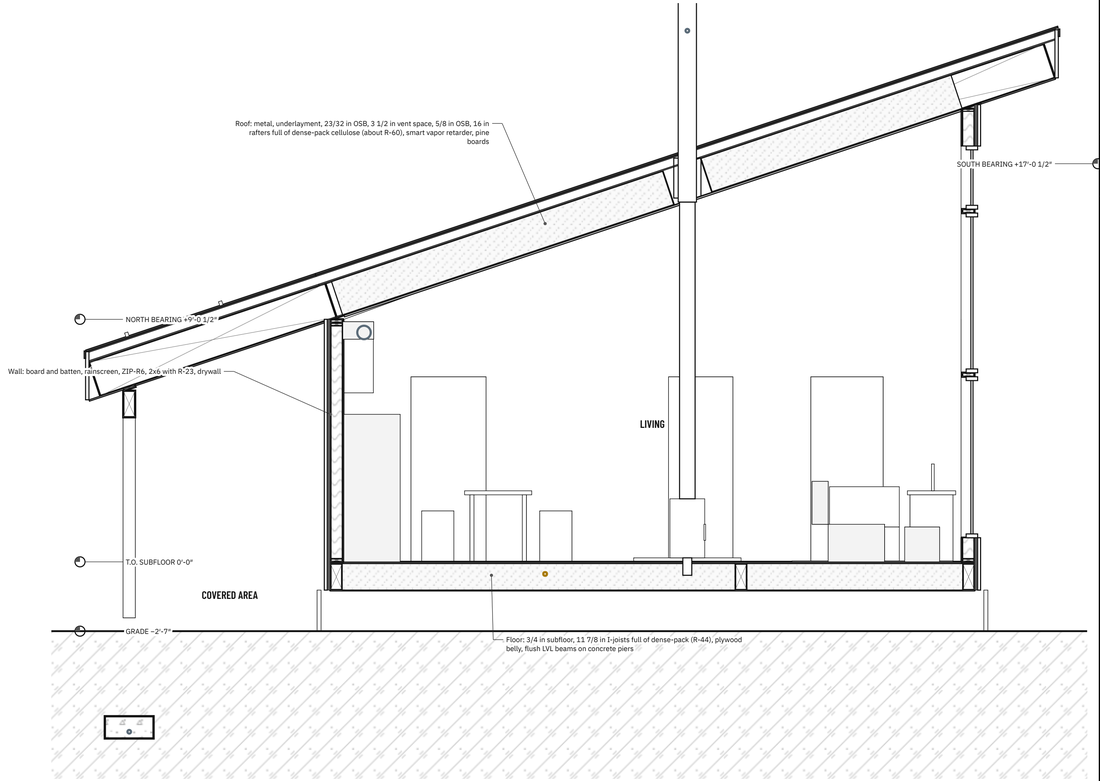
SHEET

A-201



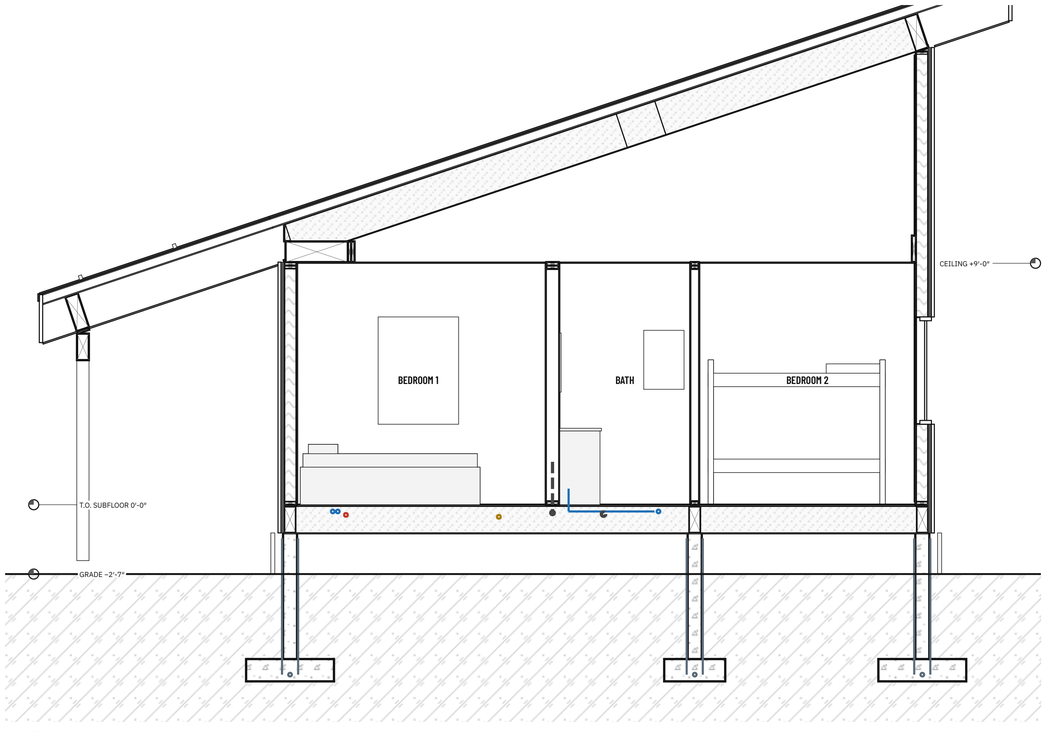
1 SECTION THROUGH LIVING ROOM AND BATH

3/8" = 1'-0" Looking north



2 SECTION THROUGH LIVING ROOM

3/8" = 1'-0" Looking east



3 SECTION THROUGH BEDROOMS AND BATH WET WALL

3/8" = 1'-0" Looking east

SECTION NOTES

1. Sections are cut straight from the 3D model, so every stud, rafter and pipe the plane passes through is shown where it falls.
2. Ceilings: living room follows the roof (9 ft-2 in to 16 ft-10 in). Bedrooms and bath 9 ft-0 in on 2x10 joists, which can carry a storage loft.
3. Roof snow load 100 psf (assumed). Rafter size and spacing on S-201, sizing on S-301.
4. Bottom of footings at least 48 in below grade or the local frost depth, whichever is deeper.
5. Vent channel above the lower roof deck runs unbroken from the north soffit to the south soffit.
6. Covered area: its own rafters run from the north wall to the LVL beam on posts and a 1 ft-6 in overhang, in the plane of the house roof. Clear height about 8'-0 3/4" under the beam.

RIME CABIN

Owner-built mountain cabin

Summit County, Colorado (Breckenridge area), about 9,500 to 10,000 ft. Lot not chosen yet.

OWNER	Sander Blue
DRAWN BY	Claude (AI)
DATE	2026-10-10
SCALE	3/8" = 1'-0"

OPTIONS SHOWN

Concrete piers · 100 psf roof snow · Propane + wood · Utility power · 3 glass bays · Standing seam · No loft · Washer-dryer · Attached garage

REVISIONS

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NOT FOR CONSTRUCTION

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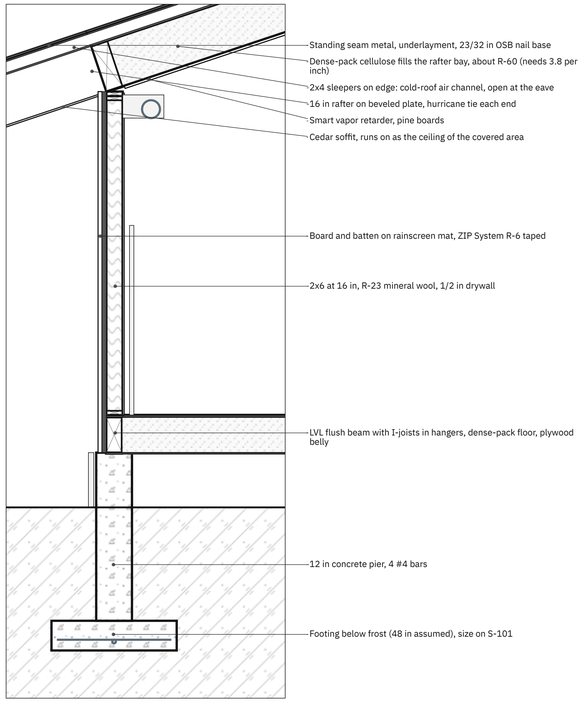
Engineer's stamp

SHEET TITLE

BUILDING SECTIONS

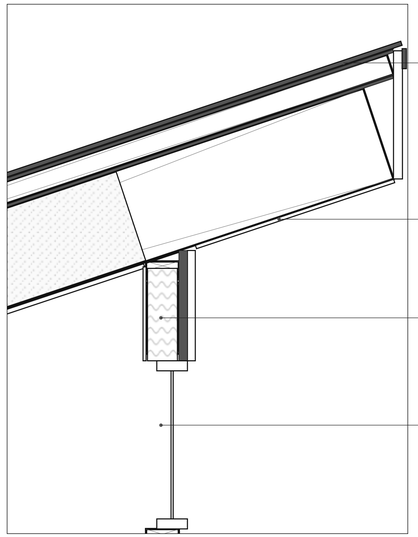
SHEET

A-301



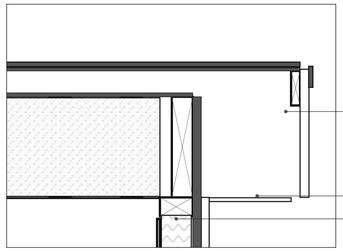
1 WALL SECTION AT NORTH WALL

1/2" = 1'-0" Cut on a pier line



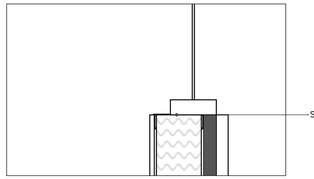
2 GLASS BAY HEAD AND HIGH EAVE

1" = 1'-0"



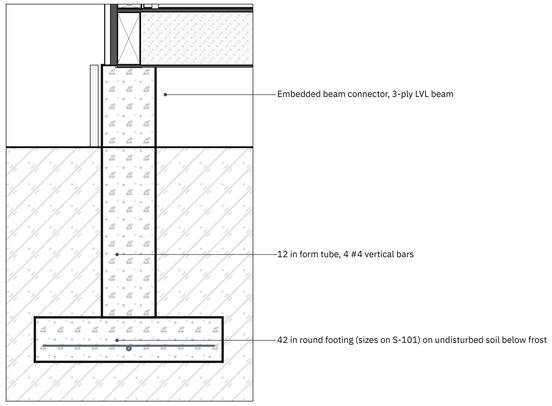
3 RAKE OVERHANG, EAST WALL

1" = 1'-0"



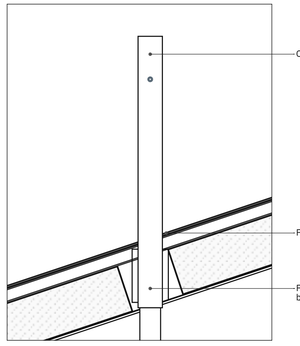
4 EGRESS WINDOW SILL

1-1/2" = 1'-0"



5 PIER AT THE NORTH BEAM

3/4" = 1'-0"



6 CHIMNEY THROUGH THE ROOF

1/2" = 1'-0"

ASSEMBLY NOTES

1. Air barrier: taped ZIP-R sheathing, taped lower roof deck, sealed subfloor, gasketed rim. Tie them together at every corner and penetration. Target 3 ACH50 or better.
2. Vapor: smart vapor retarder under the rafters. No poly in walls.
3. Windows: sloped sill pan, flashing tape shingled over the pan, head flashing over the trim, low-expansion foam and backer rod inside.
4. Penetrations: boots and collars on metal, flashing tape at the sheathing, airtight boxes inside.
5. Rainscreen mat behind all siding, insect screen at the bottom and top of the gap.
6. Details are cut from the 3D model at the marked lines and scaled. Members and connectors are planning sizes for the engineer to confirm.

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Engineer's stamp

SHEET TITLE

WALL SECTION AND DETAILS

SHEET

A-401

WINDOW AND EXTERIOR DOOR SCHEDULE

Type	Unit	Unit size	Rough opening	Qty	Operation	Sill AFF	Marks	Notes
1	Entry door, 36 in	3′-0″ x 6′-10 3/4″	3′-0 1/2″ x 6′-11 1/4″	1	Swing, in	-	W1	Insulated fiberglass with a glass lite
3	Door to the house, 36 in	3′-0″ x 6′-10 3/4″	3′-0 1/2″ x 6′-11 1/4″	1	Swing, self-closing	-	GH	1 3/4 in solid-core steel, self-closing hinges, self-latching, weatherstripped. Code-required between garage and house
A	Fixed window, 76 3/4 x 72 in	6′-4 3/4″ x 6′-0″	6′-5 1/4″ x 6′-0 1/2″	6	Fixed	0′-10 3/4″	S1a, S1b, S2a, S2b, S3a, S3b	Low-E, argon, U 0.30 or better. Two per glass bay
B	Awning window, 76 3/4 x 28 in	6′-4 3/4″ x 2′-4 1/4″	6′-5 1/4″ x 2′-4 3/4″	2	Awning	13′-1 3/4″	S1c, S3c	High vent at the top of the end bays
C	Fixed transom, 76 3/4 x 28 in	6′-4 3/4″ x 2′-4 1/4″	6′-5 1/4″ x 2′-4 3/4″	1	Fixed	13′-1 3/4″	S2c	Top of the middle bay
D	Casement, 36 x 48 in, egress	3′-0″ x 4′-0″	3′-0 1/2″ x 4′-0 1/2″	2	Casement	3′-0″	S4, E1	Bedrooms. Sill 36 in above floor
E	Casement, 36 x 36 in	3′-0″ x 3′-0″	3′-0 1/2″ x 3′-0 1/2″	1	Casement	3′-6″	N2	North wall by the table
F	Awning, 30 x 30 in	2′-6″ x 2′-6″	2′-6 1/2″ x 2′-6 1/2″	1	Awning	4′-0″	N1	Over the sink
G	Fixed window, 38 3/8 x 72 in	3′-2 3/8″ x 6′-0″	3′-2 7/8″ x 6′-0 1/2″	2	Fixed	0′-10 3/4″	W3a, W3b	Tall corner window on the west wall, stacked two high. Tempered. Half the width of a south bay unit
J	Fixed window, 38 3/8 x 20 in	3′-2 3/8″ x 1′-7 3/4″	3′-2 7/8″ x 1′-8 1/4″	1	Fixed	13′-1 3/4″	W3c	Top of the west corner window, under the sloped roof
K	Fixed window, 36 x 57 in	3′-0″ x 4′-9″	3′-0 1/2″ x 4′-9 1/2″	1	Fixed	8′-3″	W4	Clerestory over the entry door, above the porch roof

All windows low-E, argon, U-0.30 or better, dark frames. Tempered glass in the glass bays, the tall west corner window (G, J) and within 24 in of doors. Egress casements (D) give about 9 sq ft clear; confirm with the maker.

INTERIOR DOOR SCHEDULE

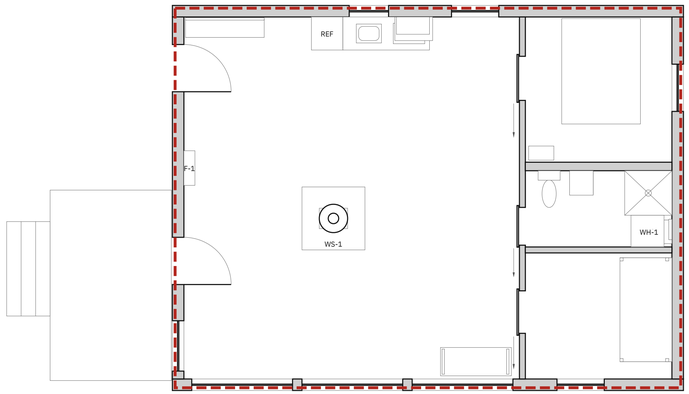
Type	Door	Size	Room	Mark
2	Sliding barn door, slab and track	2′-9 5/8″ opening, slab 2 in wider	Bedroom 1	D2
2	Sliding barn door, slab and track	2′-4 3/4″ opening, slab 2 in wider	Bath + laundry	D3
2	Sliding barn door, slab and track	2′-8 3/8″ opening, slab 2 in wider	Bedroom 2	D4

ROOM FINISH SCHEDULE

Room	Floor	Walls	Ceiling	Built-ins
Living / kitchen / entry	Rigid-core vinyl plank	Drywall, paint	Pine tongue-and-groove on the rafters	Gear bench and hooks, kitchenette, built-in desk
Bedroom 1	Rigid-core vinyl plank	Drywall, paint	Drywall, 9 ft	Queen platform bed, wardrobe
Bath + laundry	Rigid-core vinyl plank, shower pan	Moisture-resistant drywall, shower kit	Moisture-resistant drywall, 9 ft	Vanity, toilet, washer-dryer, water heater
Bedroom 2	Rigid-core vinyl plank	Drywall, paint	Drywall, 9 ft	Twin bunks

INSULATION BY ASSEMBLY

Assembly	Build-up	This design	IECC 2021, zone 7	Result
Roof	16 in rafters full of dense-pack cellulose, cold-roof vent channel above	About R-59 to R-61	R-60	Check density
Walls	2x6, R-23 mineral wool, ZIP-R6 (R-6.6 continuous)	R-23 + R-6.6 ci	R-20 + R-5 ci	Meets
Floor	11 7/8 in I-joists full of dense-pack cellulose, plywood belly	About R-44	R-38	Meets
Windows	Low-E, argon	U-0.30 or better	U-0.30	Meets
Air leakage	Taped sheathing and decks	3 ACH50 target	3 ACH50	Test



1 **AIR BARRIER LINE**

Not to scale. Red dashed line = the air barrier

1. Red dashed line: the air barrier. It is the taped ZIP-R sheathing at walls, the taped lower roof deck above, and the sealed subfloor and rim below.
2. Seal: sill gaskets, rim joints, top plates to the deck, every window and door rough opening, every pipe, wire, vent and flue penetration, the attic hatch if any.
3. Use airtight electrical boxes on outside walls. Gasket the panel and the meter penetration.
4. Blower-door test before drywall, fix leaks, then the final test.

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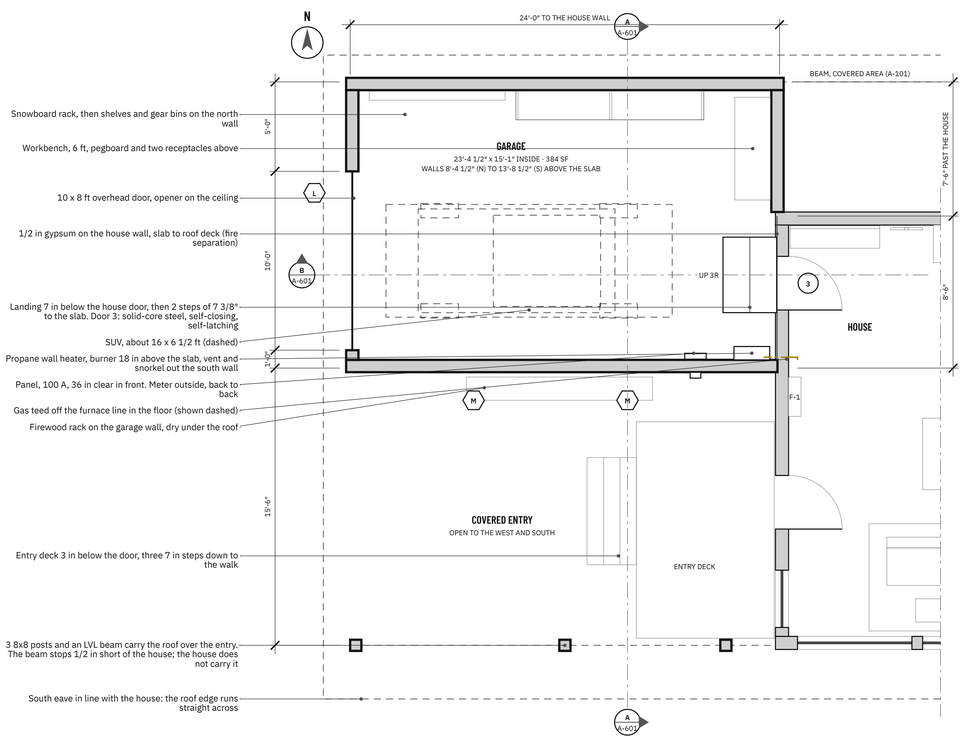
Engineer's stamp

SHEET TITLE

SCHEDULES AND ENVELOPE

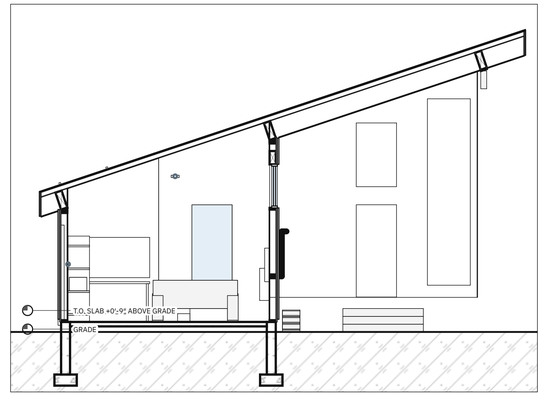
SHEET

A-501



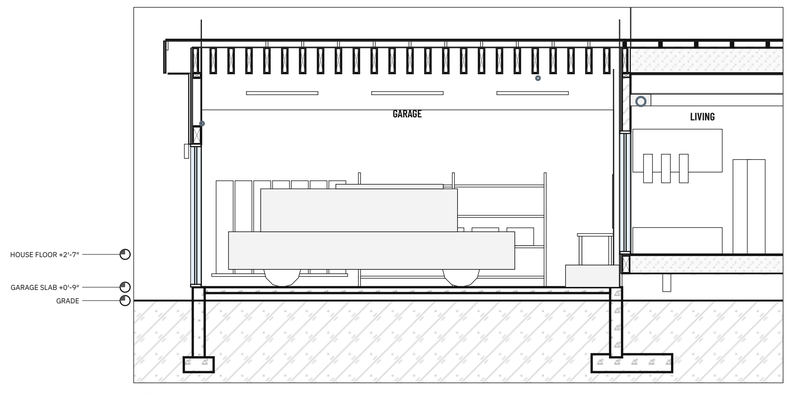
1 GARAGE AND COVERED ENTRY PLAN

1/4" = 1'-0" On the north half of the west wall, door to the west



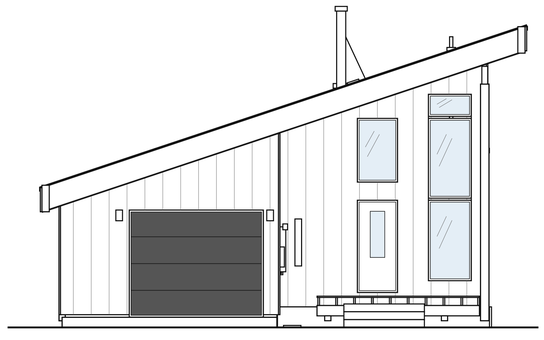
A SECTION A

3/16" = 1'-0" Looking east. Same rafters, slope and metal as the house roof



B SECTION B

1/4" = 1'-0" Looking north. One roof plane over the garage and the house



4 WEST ELEVATION

3/16" = 1'-0" The garage door and the covered entry

GARAGE DOORS AND WINDOWS

Type	Unit	Size	Qty	Sill	Marks
3	Door to the house, 36 in	3'-0" x 6'-10 3/4"	1	-	GH
L	Overhead door, 10 x 8 ft, with opener	10'-0" x 8'-0"	1	-	GD
M	Fixed window, 48 x 39 in	4'-0" x 3'-3"	4	8'-6"	GC1a, GC1b, GC2a, GC2b

GARAGE COST

Garage cost (from the parts list, my estimates)	Cost
Footing, stem wall, slab, steps	$6,180
Wall framing and connectors	$2,417
Rafters, deck, underlayment	$10,106
Metal roof, trim, soffits	$10,314
Sheathing, siding, OSB, gypsum	$3,594
Doors and windows	$4,970
Heater and gas pipe	$2,000
Lights and wiring	$561
Covered entry beam and posts	$3,010
Permit	$400
Garage subtotal, before tax and the 10% contingency	$44,536

GARAGE NOTES

1. Attached 1-car garage, 16 x 24 ft, for one SUV, a workbench and snowboard and winter gear. Same metal roof, siding and trim as the house.
2. Place: the north half of the west wall, which has no windows. It runs 7 ft-6 in past the house to the north for width. The 10 ft door faces west to the parking, under a rake, so no roof snow lands in front of it.
3. Roof: the house roof carried on west in the same plane. No step, so no drift load at the house wall and no wall flashing. The garage drops its snow north like the house, from the same eave line as the covered area east of it.
4. Fire separation: 1/2 in gypsum on the garage side of the house wall from the slab to the roof deck; seal every pipe and wire through it. Door 3 into the house: 1 3/4 in solid-core steel, self-closing, self-latching. No garage opening into a bedroom (IRC R302.5, R302.6).
5. Slab 9 in above grade, sloped 1/8 in per ft to the door, 22 in below the house floor: a concrete landing 7 in below the door, then 2 risers of 7 3/8". Frost footing and stem wall to 48 in.
6. Not insulated. ZIP sheathing (no foam) on the garage walls, OSB lining inside floor to roof. The house wall keeps its insulation and air barrier.
7. Heat: propane direct-vent wall heater, about 30,000 BTU/h, gas teed off the furnace line in the floor. Takes the chill off; it does not hold the garage warm.
8. Moved for the garage: the house panel to the garage south wall with the meter outside it; the range hood duct now runs in a soffit along the west wall and exits south of the garage. No water in the garage.
9. Power: garage circuit 13 (GFCI) for lights, opener, receptacles and the heater. Light: four clerestory windows high on the south wall.
10. Structure (S-101, S-201, S-401): frost footing and stem wall with thickened pads under the jack posts and hold-downs. LVL headers over the door and the clerestories, LVL jack posts between the windows. Garage rafters span north wall to south wall; walls braced per S-401 (OSB lining nailed as a shear wall).
11. Covered entry: separate rafters run from the garage south wall over the entry to the house's south eave line, onto an LVL beam on 3 8x8 posts in line with the south wall, each on its own pier. The beam stops 1/2 in short of the house. Open to the west and south, cedar ceiling. The entry deck, firewood, meter and vents sit under it.

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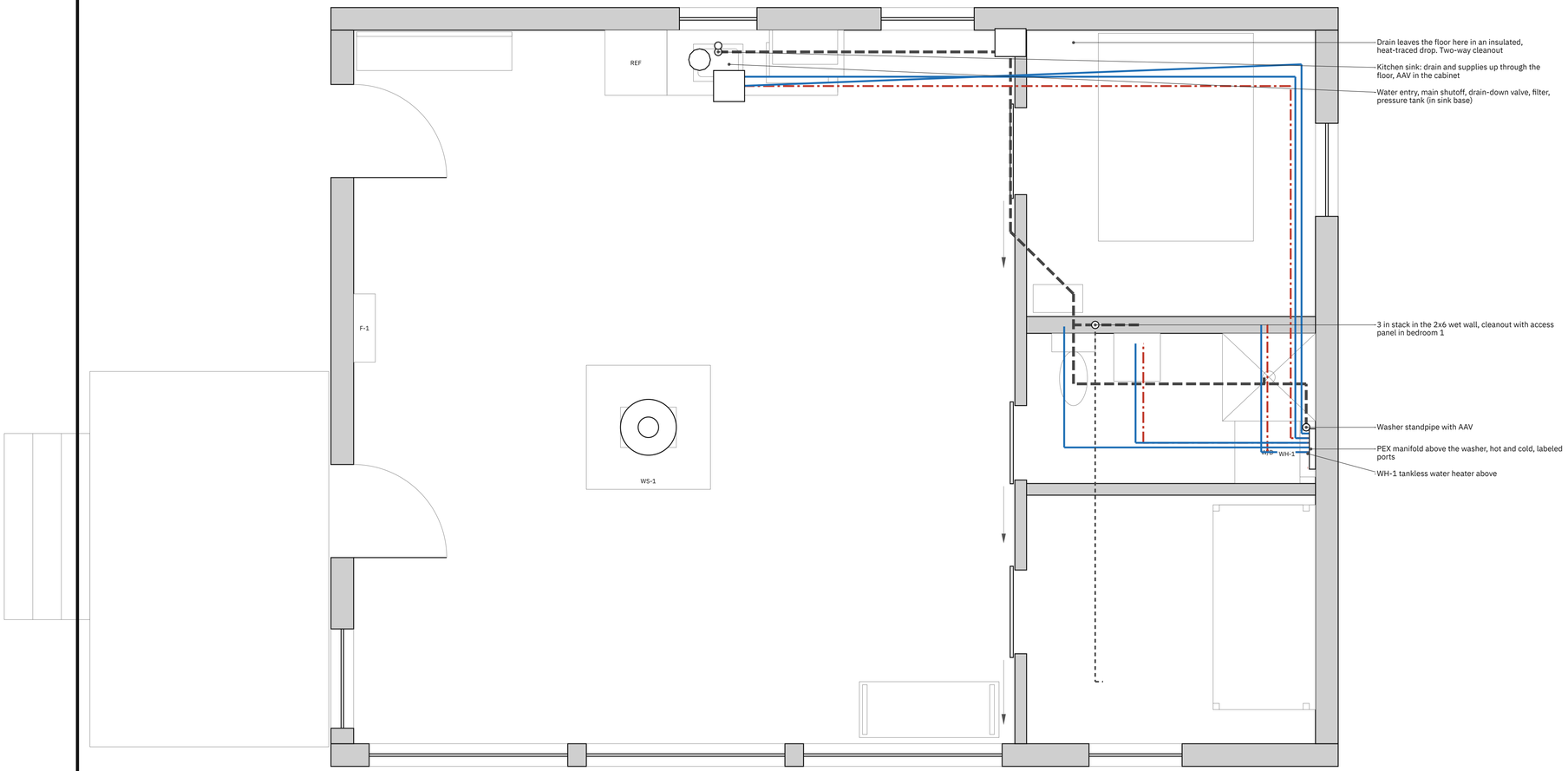
Engineer's stamp

SHEET TITLE

ATTACHED GARAGE

SHEET

A-601



1 **PLUMBING PLAN**

1/2" = 1'-0" Supply and drain lines run in the insulated floor

Cold water, PEX

Hot water, PEX

Drain, ABS or PVC

Vent

PLUMBING AND FREEZE-PROTECTION NOTES

1. No water lines in exterior walls or the attic. Supplies run in the insulated floor bays north of the middle beam, then up inside interior walls or cabinets.
2. Home-run PEX from one manifold, so each fixture has its own shutoff.
3. Drain-down: slope supplies back to the low-point drain valve at the water entry. Before a cold absence: shut the main, open the drain-down, blow out the lines, drain the water heater, put RV antifreeze in every trap and the toilet.
4. Water entry and drain drop: insulated boxes with self-regulating heat cable on a GFPE breaker (circuit 11). Water service below frost under the north snow pile.
5. Vent: 3 in stack rises in the wet wall, runs south in a ceiling joist bay and exits near the high edge of the roof, away from sliding snow. Air admittance valves at the kitchen sink and washer.
6. Test: drain lines under water or air test, supplies at 100 psi for 15 minutes, before insulation.
7. Water heater: tankless propane, direct vent through the east wall. Thermal expansion and pressure relief per the maker.

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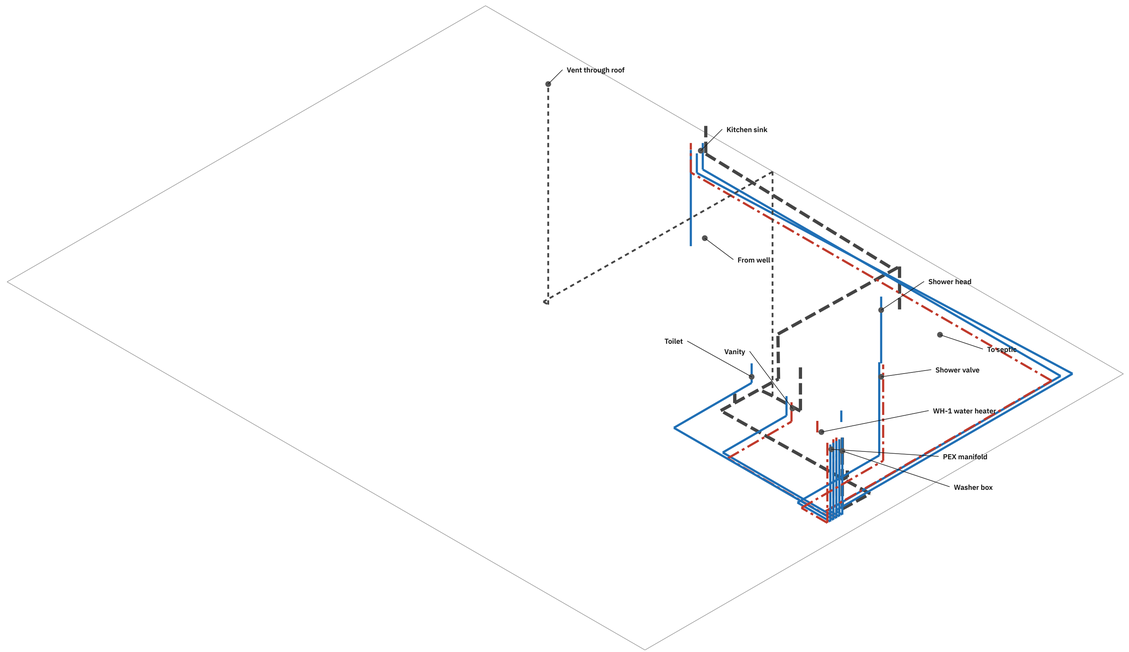
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SHEET TITLE

PLUMBING PLAN

SHEET

P-101



1 **SUPPLY AND DRAIN RISER DIAGRAM**

Not to scale. Isometric view from the southwest, drawn from the model pipe runs

Cold water
Hot water
Drain
Vent

FIXTURE SCHEDULE

Fixture	Supply	Drain	Trap	Vent	Qty
Kitchen sink	1/2 in H and C	1 1/2 in	P-trap	AAV under sink	1
Toilet, 1.28 gpf	1/2 in C	3 in	Integral	Stack	1
Vanity lavatory	1/2 in H and C	1 1/2 in	P-trap	Stack	1
Shower, 34 x 40	1/2 in H and C	2 in	P-trap	Stack (trap arm under 8 ft)	1
Washer-dryer combo	1/2 in H and C (hoses)	2 in standpipe	P-trap	AAV in box	1
Tankless water heater	3/4 in C in, 3/4 in H out	-	-	Concentric vent	1

PLUMBING MATERIALS (FROM THE PARTS LIST)

Item	Qty	Notes
PEX tubing and fittings	208 lf	Blue cold, red hot, home runs from a manifold
PEX manifold with valves	1 ea	Hot and cold, labeled ports
Drain and vent pipe	83 lf	ABS or PVC, 1 1/2 to 3 in, with fittings
Cleanout	3 ea	
Air admittance valve	2 ea	Where a vent pipe would need an outside wall
Water entry	1 lot	Frost box, heat cable, main shutoff, drain-down valve, filter
Drain drop protection	1 lot	Insulated box and heat cable where the drain leaves the floor
Pressure tank and pump control	1 ea	Small tank under the sink, constant-pressure control
Stops, washer box, test caps	1 lot	

RISER NOTES

1. Pipe lengths in the parts list come from these same runs, plus waste for fittings.
2. Building drain 3 in at 1/4 in per ft minimum. Branches 1 1/2 and 2 in at 1/4 in per ft.
3. Cleanouts at the stack base, the kitchen branch and where the drain leaves the house.
4. Shut-off at the main, at the manifold for every fixture, and at the water heater.

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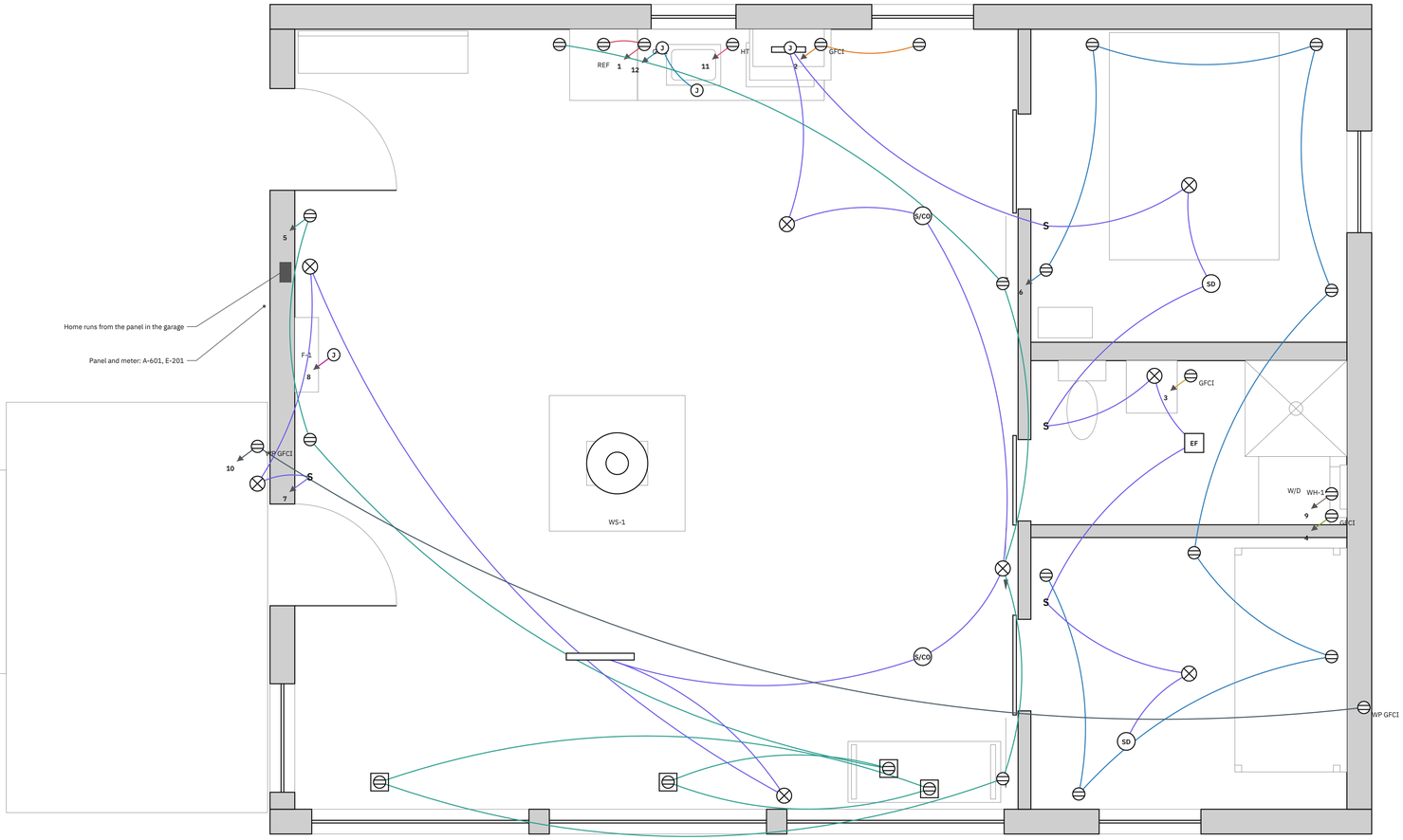
Engineer's stamp

SHEET TITLE

PLUMBING RISER DIAGRAM AND FIXTURES

SHEET

P-201



1 **ELECTRICAL PLAN**

1/2" = 1'-0" Arcs connect devices on the same circuit; arrow and number = home run to the panel

SYMBOLS

- Duplex receptacle, 16 in AFF
- GFCI receptacle
- Floor receptacle
- S — Switch, 48 in AFF
- Light fixture
- SD — Smoke alarm
- S/CO — Smoke and CO alarm
- EF — Bath fan / light
- 3 — Equipment connection

ELECTRICAL NOTES

1. Receptacles every 12 ft of wall (no point more than 6 ft from one), counters within 2 ft of each end and every 4 ft. Floor boxes where the south wall is glass.
2. GFCI: kitchen counters, bath, laundry, outdoors. AFCI: living, bedrooms and lighting. Heat cable on a GFPE breaker.
3. Smoke alarms in each bedroom; combined smoke/CO alarms in the living room near each bedroom door. Hardwired, interconnected, battery backup.
4. Outside walls: airtight boxes, gaskets at every penetration.
5. Tamper-resistant receptacles throughout. Exterior receptacle in a weatherproof in-use cover.
6. Cable lengths in the parts list come from the routed circuits in the model plus 20 percent.

RIME CABIN

Owner-built mountain cabin

Summit County, Colorado (Breckenridge area), about 9,500 to 10,000 ft. Lot not chosen yet.

OWNER	Sander Blue
DRAWN BY	Claude (AI)
DATE	2026-10-10
SCALE	1/2" = 1'-0"

OPTIONS SHOWN

Concrete piers · 100 psf roof snow · Propane + wood · Utility power · 3 glass bays · Standing seam · No loft · Washer-dryer · Attached garage

REVISIONS

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NOT FOR CONSTRUCTION

For planning and permit prep. Structural and code items need a licensed engineer and the building department.

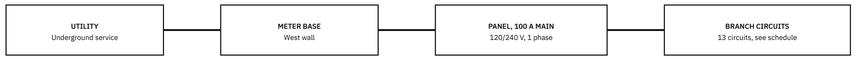
Engineer's stamp

SHEET TITLE

ELECTRICAL PLAN

SHEET

E-101



Grounding electrode: two ground rods 6 ft apart, or a concrete-encased electrode in a footing. Bond the CSST gas pipe and the water pipe.

1 ONE-LINE DIAGRAM

Utility power

PANEL SCHEDULE

Ckt	Serves	Breaker	Poles	Protection	Wire	Load VA
1	Kitchen counter A + fridge	20 A	1	GFCI/AFCI	12 AWG	1,500
2	Kitchen counter B + table	20 A	1	GFCI/AFCI	12 AWG	1,500
3	Bath receptacle	20 A	1	GFCI/AFCI	12 AWG	1,500
4	Laundry	20 A	1	GFCI/AFCI	12 AWG	1,500
5	Living room receptacles	15 A	1	AFCI	14 AWG	900
6	Bedroom receptacles	15 A	1	AFCI	14 AWG	700
7	Lights, fan, hood, alarms	15 A	1	AFCI	14 AWG	800
8	Furnace	15 A	1	AFCI	14 AWG	120
9	Water heater	15 A	1	GFCI/AFCI	14 AWG	120
10	Exterior receptacles	20 A	1	GFCI	12 AWG	360
11	Heat cable, water and drain	15 A	1	GFPE	14 AWG	200
12	Well pump (site)	20 A	2	-	12 AWG	1,800
13	Garage: lights, opener, receptacles, heater	20 A	1	GFCI	12 AWG	1,500

LOAD CALCULATION (NEC 220.82, MY CALC)

Load	VA
General lighting and receptacles, 3 VA x 768 sq ft	2,304
Small-appliance circuits, 2 x 1,500	3,000
Laundry circuit	1,500
Fixed appliances: furnace, water heater, heat cable, well pump, fridge, hood	2,690
Garage, circuit 13: lights, opener, heater blower	1,500
Total connected	10,994
First 10,000 VA at 100%	10,000
Remainder at 40%	398
Demand load	10,398
Demand current at 240 V	43 A, so a 100 A service is plenty

NOTES

1. Heating, hot water and cooking are propane, so the electrical load is small.
2. GFCI/AFCI on the schedule means one dual-function breaker. The 2020 NEC wants arc-fault protection in kitchens and laundry areas as well as ground-fault protection.
3. Utility service: confirm the service point and meter location with the utility before trenching.
4. Colorado allows homeowner electrical permits for a home you own and occupy in many counties. Check with the county; the electrician line in the parts list covers service hookup and sign-off.
5. Circuit 12 feeds the well pump; size it to the pump the driller installs.

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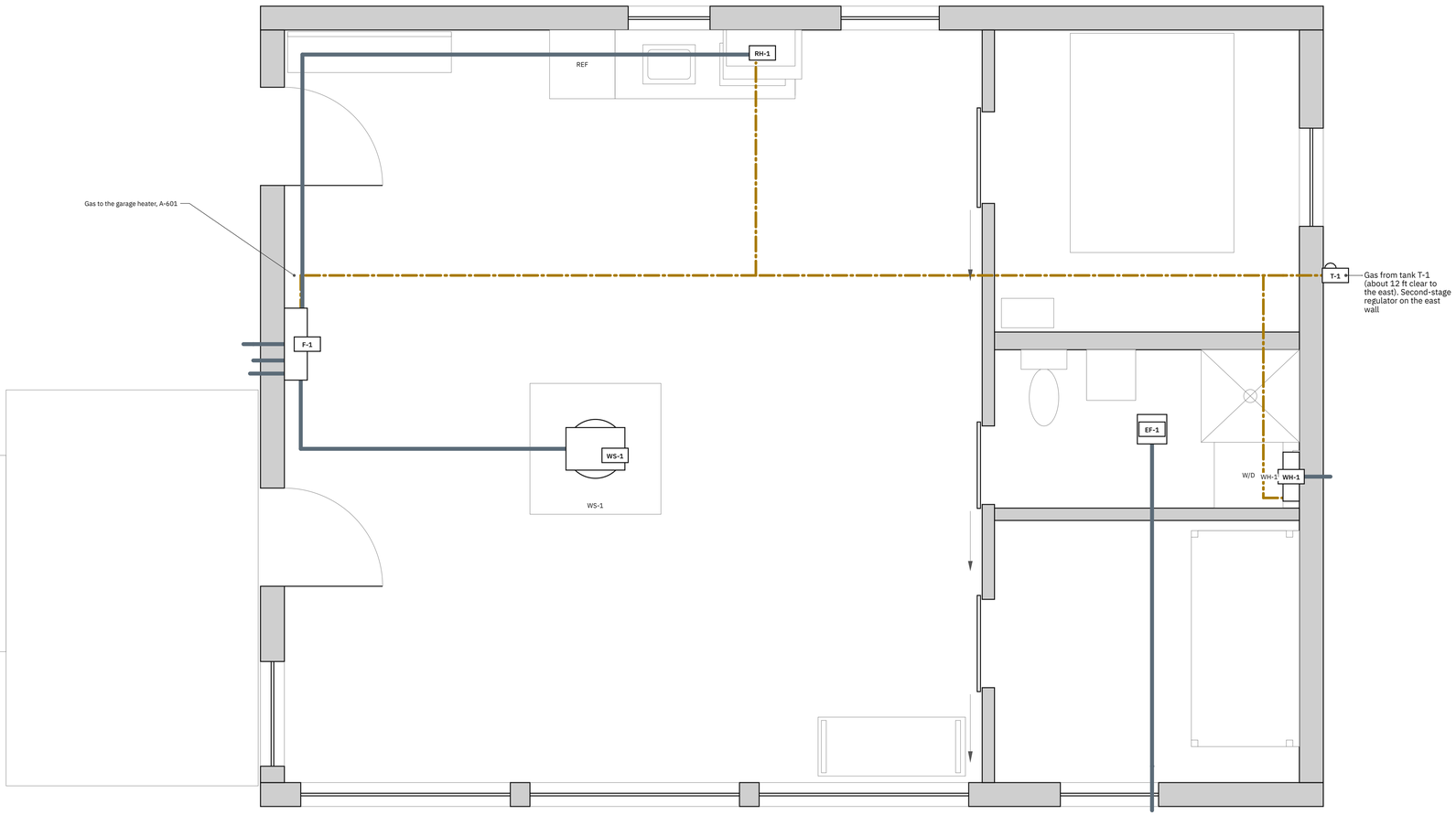
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SHEET TITLE

ONE-LINE DIAGRAM, PANEL SCHEDULE AND LOAD

SHEET

E-201



1 **MECHANICAL PLAN**

1/2" = 1'-0" Gas piping dash-dot, vents and ducts solid

EQUIPMENT SCHEDULE

Tag	Equipment	Spec	Location
F-1	Propane wall furnace	Direct vent, sealed combustion, 8,200 to 21,500 BTU/h modulating, 120 V. Snorkel vent kit lifts the terminal to about 8 ft above grade	West wall by the entry
WS-1	Wood stove	EPA certified, small firebox, outside air kit, listed clearances	Living room on a hearth pad
WH-1	Tankless water heater	Propane, indoor, concentric direct vent, about 140,000 BTU/h, 120 V	Bath, east wall above the washer
EF-1	Bath fan	50 cfm continuous, 80 cfm boost, humidity sensing. This is the whole-house ventilation fan	Bath ceiling, ducted to south wall
RH-1	Range hood	Up to 300 cfm, 6 in duct in a soffit to the west wall, backdraft damper (under 400 cfm, so no makeup air needed)	Kitchenette
T-1	Propane tank	500 gal ASME, above ground, two-stage regulation	East of the house, about 12 ft clear

VENTILATION

Ventilation	cfm
Whole-house need. 2021 IRC Table M1505.4.3(1): up to 1,500 sq ft with 0 to 1 bedrooms needs 30 cfm; 2 to 3 bedrooms needs 45 cfm	45
Bath fan EF-1, continuous	50
Bath fan boost (needs 50 intermittent or 20 continuous)	80
Range hood (under 400 cfm, so no makeup air required)	300

PROPANE LOAD

Gas load	BTU/h
Wall furnace F-1	21,500
Tankless water heater WH-1	140,000
2-burner cooktop	18,000
Garage wall heater (A-601)	30,000
Total connected	209,500
500 gal tank T-1, 30% full, 0 F (RegO D x L x K method)	315,000
Same tank at -10 F	about 157,000
A 250 gal tank at 0 F / -10 F, for comparison	197,000 / 99,000

MECHANICAL NOTES

1. Wood stove: EPA-certified, outside air kit, listed clearances, hearth pad. Class A chimney 3 ft above the roof and 2 ft above anything within 10 ft; braced; snow splitter.
2. Furnace and water heater are sealed-combustion, direct vent. The furnace vent gets a snorkel kit so it ends about 7 1/2 ft above grade; the water heater vent is 9 ft up. Both are above the assumed snow line. Clear them after big storms.
3. Range hood duct runs in a soffit over the north cabinets and down the west wall past the garage, then out the west wall under the roof: away from the snow-shed zone, 3 ft or more from windows and doors, and 3 ft above the furnace and garage heater vents.
4. Gas: CSST, bonded, pressure-tested. Tank is 500 gal because a 250 gal cannot boil off enough gas for the water heater on very cold mornings (see the table). In deep cold keep it above about 50 percent, and do not run the garage heater during a long shower.
5. Bath fan runs all the time as the house ventilation. Duct runs south in a ceiling bay to a wall cap above the bedroom 2 window.
6. Radon: the open pier space is vented. Test after move-in.

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SHEET TITLE

MECHANICAL, VENTILATION AND PROPANE

SHEET

M-101